

Does Expanding SNAP Eligibility Reduce Crime?

Evidence from Broad-Based Categorical Eligibility

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Abstract

Evidence linking food assistance to crime generally studies the amount, timing, delivery, or withdrawal of benefits among populations already eligible for assistance. The consequences of the rules determining who can enter a program remain less understood. I study this eligibility margin through Broad-Based Categorical Eligibility (BBCE), which allows states to relax SNAP asset and income screens while leaving the federal benefit formula unchanged. Combining USDA policy and participation data with agency-level FBI arrest records, I exploit staggered state adoption in a stacked difference-in-differences design. The findings indicate that BBCE increases SNAP participation by 8 percent and reduces adult arrests by 8 to 9 percent across the total, violent, and property offense indices. The results are robust across alternative comparison groups, estimators, samples, and treatment definitions. They show that eligibility design affects not only program participation but also criminal justice contact, a consequence not captured by the program's caseload or budget.

Keywords: SNAP, food assistance, eligibility rules, crime, arrests, stacked difference-in-differences

JEL: I38, K42, H53

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1 Introduction

The Supplemental Nutrition Assistance Program (SNAP) can affect crime through channels that extend beyond food consumption. By easing resource constraints, benefits may reduce financially motivated crime. By stabilizing household budgets, they may also reduce the strain that contributes to violence. Consistent with these mechanisms, existing research finds that offending responds to the amount of economic assistance (Foley, 2011; Watson et al., 2020; Palmer et al., 2019), to when SNAP benefits arrive during the month (Carr and Packham, 2019, 2021; Xu, 2026), to how they are delivered (Wright et al., 2017), and to their withdrawal (Tuttle, 2019; Deshpande and Mueller-Smith, 2022). This evidence establishes a link between economic assistance and crime. It begins, however, after eligibility has been determined.

The unresolved question lies at the program's entry point. Income limits, asset tests, and administrative requirements determine which low-income households can receive SNAP. These screens reduce take-up and shape the composition of the caseload (Currie, 2006; Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019; Homonoff and Somerville, 2021). Their consequences have nevertheless been studied mainly through enrollment, targeting, and public spending. If SNAP resources affect crime, rules that deny access may also increase criminal justice contact, imposing costs outside the program. This paper asks whether expanding SNAP eligibility reduces adult arrests for crime.

Broad-Based Categorical Eligibility (BBCE) offers a direct way to study this margin. SNAP benefits are calculated under a federal formula, but states retain discretion over important rules governing entry. BBCE allowed states to relax or eliminate the asset test and, in some cases, raise the gross income threshold. Forty-two states adopted some form of BBCE between 2000 and 2018. They adopted at different times and chose expansions of different scope. BBCE therefore generated spatial and temporal variation in access to SNAP while the federal benefit formula remained fixed.

The empirical analysis combines U.S. Department of Agriculture (USDA) policy and participation records with an agency-year panel of adult arrests constructed from the Federal Bureau of Investigation's Uniform Crime Reporting (UCR) Program. The panel covers 2,784 law enforcement agencies in 44 states from 1996 to 2016. I exploit the spatiotemporal variation in the BBCE adoption, to the effect of expanding SNAP eligibility on adult arrests for violent and property crimes. The results show that BBCE increased SNAP participation by about 8 percent and reduced adult arrests by 8 to 9 percent across the total, violent, and property indices.

Disaggregated results show that the decline is driven by rape, aggravated assault,

and larceny. The reductions span age, sex, and racial groups. Violent-crime effects are larger among adults aged 25 to 54, and index effects are larger among White than Black adults. Estimates are larger under asset-only expansions than under asset-plus-income expansions, although the differences are imprecise. The property and total results are stable across alternative comparison groups, an imputation estimator, the coding of the partial adoption year, and Poisson estimation, and an in-time placebo produces no effect. Counterfactual paths imply about 18 million additional SNAP recipient-years and 212,000 fewer adult arrests for index offenses across the sampled agencies.

The paper makes three contributions to the literature. First, it introduces the eligibility margin into research on transfers and crime. Prior studies identify effects from changes in the amount, timing, delivery, or withdrawal of assistance (Foley, 2011; Carr and Packham, 2019, 2021; Wright et al., 2017; Tuttle, 2019; Watson et al., 2020; Palmer et al., 2019; Deshpande and Mueller-Smith, 2022). These changes concern what recipients receive, among households that already qualify. BBCE instead changes who may receive SNAP, leaving the federal benefit formula untouched, and the two margins need not coincide, because admitting households that hold a car or a savings buffer extends the program to a population the existing caseload does not represent (Finkelstein and Notowidigdo, 2019). The one prior study linking BBCE to violence, Austin et al. (2023), compares states by policy category on survey-reported rates of interpersonal violence victimization and drops states that switched during its window, so it identifies a difference between state types rather than an effect of adoption. To my knowledge, this paper provides the first estimates that isolate the effect of BBCE adoption on adult arrests, recovering when the response emerges, which offenses account for it, and which groups it reaches.

Second, the paper connects research on eligibility screens and administrative burden to outcomes beyond program participation. Existing work shows that application costs, documentation requirements, and verification rules affect take-up and select who receives assistance (Currie, 2006; Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019; Homonoff and Somerville, 2021). The ordeal-based rationale for such screens is that costly procedures improve targeting by deterring applicants with lower need (Nichols and Zeckhauser, 1982). Evaluations of this tradeoff have focused largely on enrollment and caseload composition. By examining adult criminal justice contact, this paper identifies a consequence of exclusion that program-based evaluations do not measure.

Third, the paper broadens the evaluation of BBCE. Existing economic studies examine participation, caseload composition, targeting, and program cost (Klerman and Danielson, 2011; Ganong and Liebman, 2018; Wang, 2026; Han and Wu, 2026). This paper extends that evidence to adult arrests for violent and property crimes. It also uses the un-

derlying arrest returns to study variation by age, sex, and race, distinguishes the asset and income policy levers, and translates the estimates into aggregate policy quantities. BBCE is therefore evaluated not only by whom it admits and what it costs, but also by outcomes borne outside the program.

The remainder of the paper proceeds as follows. Section 2 gives the institutional background, Section 3 the conceptual framework, Section 4 the data, and Section 5 the empirical strategy. Section 6 reports the results, Section 7 the robustness checks, and Section 8 the discussion. Section 9 concludes.

2 Institutional Background

SNAP is the largest nutrition assistance program in the United States. In fiscal year 2025, it served an average of 42.1 million people each month, and federal spending totaled \$101.7 billion (USDA Economic Research Service, 2026). During the period studied here, the federal government financed benefits, while the federal and state governments shared administrative costs. States administered the program, but federal law established the baseline eligibility rules and the schedule determining benefits.

Under the regular federal pathway, a household generally must pass three financial tests. Gross monthly income cannot exceed 130 percent of the federal poverty guideline, and net income after allowable deductions cannot exceed 100 percent. Households containing an elderly or disabled member are exempt from the gross-income test but remain subject to the net-income test. A household must also hold countable resources below the federal asset limit. During most of the study period, that limit was \$2,000 for ordinary households and \$3,000 for households containing an elderly or disabled member. Congress indexed the limits after 2008 (Oliveira et al., 2018; USDA Food and Nutrition Service, 2026). A primary residence and specified retirement and education accounts are excluded, while cash, funds in financial accounts, and some vehicle value may be counted. Separate rules govern citizenship, student status, and work requirements, and BBCE does not waive them.

Households may also enter SNAP through categorical eligibility. Under the traditional pathway, a household receiving cash assistance from programs such as Temporary Assistance for Needy Families (TANF), Supplemental Security Income, or state General Assistance is deemed to satisfy the SNAP gross-income and asset tests. The policy avoids repeating financial tests already conducted by another means-tested program (Oliveira et al., 2018).

The Personal Responsibility and Work Opportunity Reconciliation Act of 1996 re-

placed Aid to Families with Dependent Children with the TANF block grant and gave states greater discretion over TANF-funded benefits and services. USDA regulations issued in 2000 clarified that receipt of, or authorization to receive, a noncash TANF or state maintenance-of-effort benefit could also confer categorical eligibility for SNAP. Qualifying services could include an information brochure, referral, or hotline rather than a cash payment (U.S. Government Accountability Office, 2007; Oliveira et al., 2018). This broader use of categorical eligibility became known as Broad-Based Categorical Eligibility.

BBCE gives states two principal financial levers. A state may raise or eliminate the asset limit, and it may raise the gross-income threshold above the federal level of 130 percent, up to 200 percent of the federal poverty guideline. States that raised the gross-income threshold generally also removed the formal net-income eligibility test. The benefit formula nevertheless creates an effective net-income ceiling because the allotment approaches zero as net income rises. Households entering through BBCE must still apply, satisfy the remaining SNAP rules, and have net income low enough to receive a positive allotment (U.S. Government Accountability Office, 2012; Han and Wu, 2026; Wang, 2026). BBCE therefore expands the set of households that may qualify, but it does not increase the maximum allotment or change the 30-percent benefit reduction rate.

Eligibility and benefit determination are separate, which is what makes BBCE a clean eligibility experiment. Conditional on eligibility, the monthly allotment for household h in month t is

$$B_{ht} = \bar{B}_{h,t} - 0.30Y_{ht}^N, \quad (1)$$

where $\bar{B}_{h,t}$ is the maximum allotment for the household's size and Y_{ht}^N is net income. The maximum allotment is based on the USDA Thrifty Food Plan and is updated annually. Net income equals gross income minus allowable deductions for earned income and expenses such as dependent care, medical care, and shelter. A household with zero net income receives the maximum allotment, and the allotment falls by 30 cents for each additional dollar of net income, subject to the statutory minimum for small households (Oliveira et al., 2018; Han and Wu, 2026). Assets do not enter this formula, so BBCE changes who may receive SNAP without changing what an eligible household receives.

Figure 1 summarizes BBCE adoption between 1996 and 2019. Forty-two states, including the District of Columbia, adopted BBCE between 2000 and 2018, while nine did not adopt by the end of 2019. Eleven states had adopted by the end of 2004. Adoption then accelerated around the Great Recession, with 28 states adopting between 2008 and 2011. No additional state adopted from 2012 through 2017, and Indiana adopted in 2018.

USDA encouraged the policy during the recession as a way to reduce resource verification, administrative time, and payment errors as SNAP caseloads increased (USDA Food and Nutrition Service, 2009; Klerman and Danielson, 2011; Ganong and Liebman, 2018; Han and Wu, 2026; Wang, 2026).

[Figure 1 about here]

The policy also varied in scope at initial adoption. Eighteen states relaxed the asset test without raising the general gross-income threshold. Twenty-four relaxed the asset test and raised the gross-income threshold, most commonly to 185 or 200 percent of the federal poverty guideline. Montana and North Dakota belong to this second descriptive category because they changed both thresholds, but they retained the net-income test. Following Han and Wu (2026), the lever analysis excludes these two incomplete income reforms and distinguishes asset-only expansions from expansions that raised the gross-income threshold and eliminated the net-income test.

The totals in Figure 1 describe the national policy history rather than the estimation sample. The empirical design retains 32 adoption events whose full event windows and agency reporting histories fit within the analysis period. Section 4 describes that sample restriction and the coding of treatment dates and policy scope.

3 Conceptual Framework

BBCE changes an eligibility rule rather than the federal benefit schedule. Its effect on arrests therefore depends on two links. The policy must first change who participates in SNAP, and the resulting change in program access must then alter household circumstances enough to affect offending or the probability that an offense produces an arrest. This section organizes those links and states the empirical predictions. The framework is deliberately reduced-form. The data do not link SNAP recipients to arrestees, so the analysis identifies the effect of the policy on arrests in treated jurisdictions rather than a treatment effect on newly enrolled households.

3.1 Eligibility, take-up, and household resources

Let $\mathcal{E}_{st}^0$ denote households eligible under the regular federal rules in state s and year t , and let $\mathcal{M}_{st}$ denote households made newly eligible by BBCE. If $p_h(d)$ is household h 's probability of participating under eligibility regime $d \in \{0, 1\}$, the policy-induced change in enrollment can be written as

$$\Delta N_{st} = \int_{\mathcal{M}_{st}} p_h(1) dh + \int_{\mathcal{E}_{st}^0} [p_h(1) - p_h(0)] dh. \quad (2)$$

The first term is enrollment among households that could not qualify before BBCE. The second is an inframarginal response among households already eligible under federal rules. That response can arise if BBCE reduces verification burdens, corrects mistaken beliefs about eligibility, or prevents households from leaving SNAP when earnings fluctuate around the federal threshold. Application and recertification frictions can therefore make an eligibility reform increase participation even among households whose statutory eligibility did not change (Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019; Homonoff and Somerville, 2021; Wang, 2026; Han and Wu, 2026).

For every household induced to participate, SNAP adds food resources according to the unchanged federal formula in equation (1). Because food would otherwise be purchased with cash, the transfer also frees resources for other necessities. The first stage is consequently part of the mechanism, not merely a validation exercise. It establishes whether changing the screen changed the resources reaching households.

Equation (2) also clarifies why the effect of BBCE cannot be inferred from studies of benefit generosity. A larger benefit for an existing recipient changes the intensive margin. BBCE changes the extensive margin and may also change take-up among the already eligible. The households affected on these margins can differ in income, assets, family structure, and exposure to the criminal justice system. The reduced-form effect of BBCE thus combines how many households enter or remain enrolled, which households they are, and how each responds to SNAP.

3.2 From program access to arrests

The economic model of crime begins with a comparison between the returns to legal and illegal activity (Becker, 1968; Ehrlich, 1973). SNAP access can shift that comparison through three related channels.

The first is a resource channel. Greater food purchasing power lowers the marginal value of resources obtained illegally and can reduce the need to meet short-run material gaps through acquisitive crime. This mechanism gives its clearest prediction for common property offenses, especially larceny and motor vehicle theft. Evidence that crime varies over the SNAP benefit month and with the timing or form of transfers supports this interpretation (Foley, 2011; Carr and Packham, 2019; Watson et al., 2020; Wright et al., 2017).

The second is a household-stability channel. Food insecurity and unstable finances

can generate stress and conflict even when material gain is not the object of an offense. More reliable access to food may reduce these pressures and thereby reduce interpersonal violence. This mechanism points most directly to assault rather than property crime. It can also extend beyond the person who applies for SNAP because eligibility, benefits, and consumption are determined at the household level. Evidence that domestic violence rises as SNAP benefits are exhausted within the month and that household income shocks affect later criminal involvement is consistent with such spillovers (Akee et al., 2010; Carr and Packham, 2021; Deshpande and Mueller-Smith, 2022).

The third is an economic-stability channel that is especially relevant to the design of BBCE. Relaxing the asset test permits households to retain modest savings and, where vehicle rules bind, transportation that supports employment. Raising the gross-income threshold can reduce eligibility losses caused by temporary earnings increases. Both changes may improve job continuity and a household's ability to absorb shocks, raising the opportunity cost of crime. These responses should develop more slowly than relief from an immediate resource shortage and need not be concentrated in a single offense category. Asset and income screens can also affect saving and work incentives in the opposite direction, however, so this channel does not deliver an unambiguous sign (Sullivan, 2006; Raphael and Winter-Ebmer, 2001; Gould et al., 2002; Han and Wu, 2026).

These mechanisms are not mutually exclusive. The same eligibility expansion can reduce property arrests through resources and violent arrests through household stability. Nor do they imply that every newly admitted household is at risk of offending. A small response among households near that margin, or a spillover to other household members, can change aggregate arrests without describing the typical SNAP recipient.

3.3 Empirical implications

The framework yields four implications for the empirical analysis. First, an effect operating through SNAP requires a participation response. Because households learn about eligibility and encounter the new rules at application or recertification, participation and arrest responses may build after adoption rather than appear fully in the first year. Neither should precede the policy.

Second, offense composition helps distinguish channels. A response concentrated in larceny or motor vehicle theft is most consistent with the resource channel, while a response in aggravated assault is consistent with reduced household strain. Similar movements in the violent and property indices would suggest that no single channel is sufficient. The total index summarizes the overall effect, while its two components show

where it arises.

Third, the policy levers change both reach and composition. Asset-only reforms principally admit income-eligible households with savings or vehicles. Reforms that also raise the income threshold reach a broader population and can produce larger marginal and inframarginal enrollment responses (Han and Wu, 2026). A larger arrest coefficient under one lever could therefore reflect a larger first stage, a different population, or a different response per participating household. The lever estimates are informative about where the reduced-form effect originates, but they do not by themselves rank the two rules or recover effects per newly enrolled household.

Fourth, demographic differences are informative but not decisive mechanism tests. Young adults have the highest baseline propensity to be arrested, while prime-age adults are more likely to apply for SNAP on behalf of a household and to face its income and asset screens. Effects by sex may reflect differences in baseline offending, household roles, or exposure to interpersonal conflict. Effects by race may additionally reflect differences in eligibility, take-up, adoption geography, and the probability that an offense results in arrest. Without demographic-specific participation first stages or linked household records, these decompositions should be interpreted as heterogeneity in the policy's reduced-form effect rather than as evidence that any one channel has been isolated.

The sign and size of the arrest response are therefore not mechanical. BBCE can increase household resources and stability, but it reaches households that differ from existing recipients and may alter work and saving incentives. The empirical analysis tests whether the net effect is detectable, when it emerges, and whether its composition accords with the channels described above.

4 Data

I assemble the analysis from three principal sources. The FBI Uniform Crime Reporting Program provides agency-level arrest records. The USDA Economic Research Service (ERS) SNAP Policy Database identifies BBCE adoption and other state SNAP rules, while the University of Kentucky Center for Poverty Research (UKCPR) National Welfare Data provide SNAP participation and state economic conditions. I also use the FBI's Return A and Law Enforcement Officers Killed and Assaulted collections to examine how arrests relate to reported offenses and police staffing. Appendix Table A7 provides a variable-level source crosswalk.

4.1 Arrest measures and analysis sample

The outcome is adult arrests. I use arrest records because the FBI reports the arrestee's age, sex, and race, which identifies the demographic groups accounting for changes in aggregate arrests, and because validation studies find that the UCR provides credible measures of serious index crimes (Gove et al., 1985; Nagin, 1998). An arrest is recorded only after an enforcement decision, so a decline could in principle reflect policing rather than offending.¹

I build the arrest panel from the FBI's Arrests by Age, Sex, and Race files distributed by the Inter-university Consortium for Political and Social Research (ICPSR) (Federal Bureau of Investigation, 1996–2019). Each agency-month-offense record reports arrests across fourteen age-by-sex cells and four race cells, along with the population covered by the agency. I pool the 1996–2019 releases, remove months the FBI designates as invalid, harmonize offense and reporting codes across years, and aggregate 32.7 million records to the agency-year level. I define an agency's covered population as the largest monthly population reported during the year.

I construct adult arrests directly by summing the twelve age-by-sex cells for persons age eighteen and older. The violent index combines murder and nonnegligent manslaughter, rape, robbery, and aggravated assault. The property index combines burglary, larceny, and motor vehicle theft, while the total index sums the two. I exclude arson because its reporting coverage is incomplete. I also construct outcomes for adults ages 18–24, 25–54, and 55 and over, men and women, and White and Black arrestees. These are aggregated reporting cells rather than individual records, so I cannot link an arrestee to SNAP receipt. Appendix A5 reports the FBI codes and definitions underlying each outcome. The estimating outcome is the log of adult arrests per 100,000 covered residents, and baseline specifications are weighted by covered population.

Because UCR participation is voluntary, incomplete reporting can appear as a change in arrests. I address this problem by requiring each agency to report all twelve months in every year of a stack's event window (Maltz and Targonski, 2002; Dave et al., 2021). I also require its mean covered population over that window to reach 10,000 (Corman et al., 2017). I apply both restrictions within each stack and hold its agency roster fixed throughout the event window, preventing reporting gaps or changes in sample composition from driving the estimates. The baseline covers 1996–2016.² The final sample contains 2,784

¹Section 7.7 tests this. Sworn staffing falls by 3.8 log points after adoption, the measured elasticity of arrests with respect to sworn officers is 0.06 to 0.10, and conditioning on staffing moves the total-index estimate by 0.003 log points.

²The 2017 release breaks the age-by-sex reporting the outcome is built from as departments migrate to the National Incident-Based Reporting System, a transition in the FBI's collection system unrelated to state

agencies in 44 states. Because comparison agencies can enter multiple stacks, the estimation sample contains 255,640 stack-agency-year observations across 32 adoption events. Section 7.6 shows that removing the population floor, or raising it to 25,000 or 50,000, produces little change in the estimates, and that the reporting rule matters more than the floor does.

4.2 BBCE treatment

Information on BBCE comes from the USDA Economic Research Service SNAP Policy Database (USDA Economic Research Service, 2024). The database records state eligibility rules by month beginning in 1996, including whether BBCE was in force, the applicable asset rule, the gross-income threshold, and the net-income test. Monthly frequency matters because states often implemented BBCE during, rather than at the beginning of, a calendar year.

I define implementation as the first month in which the database records BBCE as active. A state's adoption cohort is the calendar year containing that month. Because arrests are measured annually, the main treatment indicator equals one for agencies in the adopting state beginning in the implementation year, including a potentially partial first year, and remains one thereafter. Thirty-eight of the forty-two adopting states implemented partway through a calendar year, so this choice attenuates the first-year coefficient toward zero rather than inflating it. The underlying monthly records are retained to identify exposure during the implementation year and subsequent policy reversals.

Forty-two states adopted BBCE by 2019. Thirty-two enter the baseline arrest analysis, the cohorts whose event windows fit inside 1996 through 2016. Their reforms differed in scope. Some states changed only the asset test, while others also raised the gross-income threshold and eliminated the net-income test. Following Han and Wu (2026), I classify states according to the rules introduced at initial adoption. The sample contains fifteen asset-only reforms and seventeen asset-plus-income reforms. Montana and North Dakota are excluded because their initial expansions raised the gross-income threshold while retaining the net-income test, preventing a clean assignment to either group.

The main treatment remains absorbing so that it measures entry into the BBCE regime. Louisiana adopted BBCE in 2010 and reversed it in 2014. Mississippi's 2019 reversal falls outside the baseline period. The robustness analysis drops the Louisiana stack entirely rather than redefining the main treatment.³

policy.

³Excluding Louisiana leaves every estimate within one standard error of the reported values. Any bias from the reversal runs toward zero, so the reported estimates are if anything conservative.

4.3 Participation and controls

State-year SNAP recipient and household caseload counts come from the UKCPR National Welfare Data (University of Kentucky Center for Poverty Research, 2023). UKCPR compiles these measures from USDA Food and Nutrition Service administrative records. Recipient and household counts enter in logs, and recipient participation is also expressed relative to state population.

The analysis includes thirteen state-year controls. Seven come from UKCPR: the unemployment rate, poverty rate, log minimum wage, log maximum TANF benefit for a family of three, the state Earned Income Tax Credit (EITC) rate, an indicator for a Democratic governor, and log state population. Six additional controls come from the SNAP Policy Database: simplified reporting, combined application projects, vehicle exclusions, Electronic Benefit Transfer (EBT) issuance, online applications, and short recertification periods. These variables capture economic conditions, related income-support policies, and contemporaneous changes in SNAP administration.

5 Empirical Strategy

5.1 Stacked difference-in-differences

I estimate a stacked difference-in-differences design following Cengiz et al. (2019), Deshpande and Li (2019), and Wing et al. (2024). This design fits the setting because states adopted BBCE at different times and the policy’s effects may evolve with exposure. A conventional two-way fixed effects estimator can use already-treated states as controls for later adopters, contaminating the comparison when treatment effects are dynamic (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfœuille, 2020; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021). I avoid these comparisons by organizing the data around individual adoption events and requiring every comparison state to remain untreated throughout the relevant event window.⁴

For each adoption event, I construct a separate stack covering four years before adoption and six years after, $k = t - g_s \in [-4, +6]$, where g_s is the adoption year of the focal state in stack s . Each stack contains agencies in the focal state and agencies in states that either never adopt BBCE during the sample period or adopt after $g_s + 6$. The latter condition allows not-yet-treated states to serve as comparisons while ensuring that none becomes treated during the focal state’s window. I apply the reporting and population

⁴Appendix A2 reports the two-way fixed effects benchmark and its decomposition. Comparisons using already-treated states receive 29 percent of the weight and pull every index estimate toward zero.

requirements described in Section 4.1 within each stack and hold its agency roster fixed across the window. Thirty-two adoption events satisfy these requirements. Appendix A4 documents the construction and comparison pool for every stack.

The stacks are pooled and estimated jointly,

$$y_{ist} = \alpha_{si} + \lambda_{st} + \gamma D_{ist} + X'_{it}\beta + \varepsilon_{ist}, \quad (3)$$

where i indexes agencies, s stacks, and t calendar years. The outcome y_{ist} is the logged adult arrest rate, and X_{it} contains the state-year controls assigned to agency i . The treatment indicator D_{ist} equals one for agencies in the focal state beginning in its BBCE adoption year.

The stack-by-agency fixed effects, α_{si} , absorb permanent differences between agencies within each comparison. The stack-by-year fixed effects, λ_{st} , absorb shocks shared by the treated and comparison agencies in a given stack and calendar year, including the Great Recession and the longer national decline in crime. An agency used in several stacks receives a separate fixed effect in each. Identification therefore comes from comparing the change within a focal state's agencies with the contemporaneous change among its own untreated comparison agencies. The coefficient γ is the covered-population-weighted average post-adoption effect across the 32 stacks.

To trace the dynamics of the response I estimate

$$y_{ist} = \alpha_{si} + \lambda_{st} + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - g_s = k\} \cdot \text{Treated}_{is} + X'_{it}\beta + \varepsilon_{ist}, \quad (4)$$

where the summation covers $k \in [-4, +6]$ and the year before adoption, $k = -1$, is omitted. The lead coefficients reveal whether treated and comparison agencies were moving similarly before adoption, while the seven lag coefficients trace the post-adoption response. I report their average as a summary of the dynamic effect, but retain $\hat{\gamma}$ from equation (3) as the primary estimate.

I weight observations by agency covered population, so the estimates describe the average effect for residents represented by the sample agencies rather than the average agency.⁵ I cluster standard errors by state. BBCE is assigned at the state level and a state's agencies share its policy environment and its reporting practices, so their residuals

⁵Solon et al. (2015) emphasize that weighting in causal estimation should be guided by the estimand. I weight observations by agency covered population so that the estimates describe the average effect for residents represented by the sample rather than the average agency. Panel B of Table 8 removes these weights. The static estimates are somewhat smaller but remain negative and statistically significant across all three indices, although the corresponding unweighted event-study estimates are more attenuated.

are correlated within a state and independent across states. This produces 44 clusters, which is enough for the usual cluster-robust asymptotics to be a reasonable approximation (Cameron and Miller, 2015).

I designate the total, violent, and property indices as the primary outcomes. The component offenses are the FBI's fixed Part I taxonomy and are exhaustive within each index, so they describe the composition of an effect the indices establish.

5.2 Identifying assumption

Identification requires that, absent BBCE, arrests in adopting states would have evolved like arrests in their stack-specific comparison agencies. The design cannot test this assumption directly, but it has testable implications: no divergence in the years before adoption, no effect when adoption is assigned to a period in which it did not occur, no relationship between adoption timing and pre-period state conditions, and no sensitivity to the contemporaneous policy and macroeconomic changes that accompanied adoption. Section 7 takes up each in turn.

6 Results

6.1 Descriptive statistics

Table 1 reports population-weighted means and standard deviations for the full estimation sample and separately for agencies in the 32 adopting and 12 comparison states. The sample averages 608 adult index arrests per 100,000 covered residents. Of these, approximately 198 are violent arrests and 411 are property arrests. Aggravated assault is the most common violent offense, at 152 arrests per 100,000, while larceny dominates the property index, at 302 arrests per 100,000.

[Table 1 about here]

The two groups differ in levels. Agencies in adopting states average 210 violent arrests per 100,000, compared with 154 in comparison states, but fewer property arrests, 396 compared with 466. Adopting-state agencies also cover larger populations, averaging approximately 449,000 residents compared with 179,000. Their states have higher unemployment rates, 6.4 versus 5.3 percent, and higher poverty rates, 13.4 versus 11.9 percent. These contrasts describe the composition of the sample rather than the effect of

BBCE. The stacked specification absorbs persistent agency differences and identifies the effect from changes within each comparison.

Figure 2 describes how arrests evolved for adults ages 18–24 and 25–54 on a roster of agencies that report complete data in every year from 1996 through 2016. A balanced roster is important because the number of agencies represented in the stack-based estimation sample varies across calendar years, and using that changing roster would mix movements in arrests with changes in sample composition.

[Figure 2 about here]

Arrests decline over the sample period for both age groups and across all three indices. Adults ages 25–54 account for between 1.3 and 2.5 times as many arrests as adults ages 18–24 when both are expressed per 100,000 covered residents. Because the older category spans a wider age range, this comparison describes each group’s contribution to aggregate arrests rather than its individual arrest risk. Section 6.5 estimates the effects separately by age.

6.2 BBCE and SNAP participation

I first examine whether BBCE increased SNAP participation. Table 2 applies equation (3) to a state-year panel constructed from the same 32 adoption events used in the arrest analysis. The specifications include stack-by-state and stack-by-year fixed effects, the full set of state controls, and state-population weights.

[Table 2 about here]

BBCE increased the number of SNAP recipients by 7.8 percent and household caseloads by 9.1 percent. The unweighted estimates are slightly larger, at 8.8 and 10.0 percent, and yield the same conclusion. Recipient counts and recipients per capita produce the same treatment coefficient because log state population enters the specification as a control.

Figure 3 shows how participation evolved around adoption. The pre-adoption coefficients remain close to zero, and the joint tests do not reject that they are jointly zero, with p -values of 0.391 for recipients and 0.854 for household caseloads. Participation begins increasing in the adoption year and continues rising through the sixth post-adoption year, with no indication that the response has fully levelled off by the end of the window. The static and dynamic estimates establish a clear first stage between BBCE adoption and SNAP participation.

[Figure 3 about here]

6.3 Arrests

Table 3 reports the pooled effects of BBCE on the three arrest indices and their component offenses. The three headline estimates are close together. BBCE reduces the total index by 8.7 log points, the violent index by 7.9 log points, and the property index by 9.2 log points. These coefficients correspond to declines of 8.3, 7.6, and 8.8 percent. All three are statistically distinguishable from zero at the 5 percent level, with p -values of 0.045, 0.029, and 0.044.

[Table 3 about here]

The similar proportional effects conceal differences in absolute magnitude. Applying the estimates to mean arrest rates in adopting states implies approximately 16 fewer violent arrests and 35 fewer property arrests per 100,000 covered residents each year, or 50 fewer index arrests in total. Property offenses account for roughly two thirds of this decline because they occur more frequently at baseline.

Within the violent index, the decline is concentrated in rape and aggravated assault, which fall by 12.5 and 9.4 log points. Aggravated assault represents 77 percent of violent arrests at baseline and accounts for most of the estimated decline in the violent index. Rape falls by a larger proportion but contributes under one arrest per 100,000 because its baseline rate is low. Murder and robbery show no reliable changes.

Within the property index, larceny falls by 10.3 log points and motor vehicle theft by 11.8 log points, though the latter is imprecise. Motor vehicle theft has the largest proportional point estimate, but larceny accounts for most of the absolute change, approximately 28 of the 35 fewer property arrests per 100,000. The burglary coefficient is negative in the static specification, but its pre-adoption coefficients are jointly significant and its average post-adoption event-study estimate is close to zero. I therefore treat the burglary result as a null.

Figure 4 traces the estimates over event time. The joint lead tests do not reject parallel pre-adoption paths for the total or property indices, with p -values of 0.254 and 0.383; the violent leads are marginal at 0.076. All three series turn negative in the adoption year, reach their most negative values between the first and third post-adoption years, and then move back toward zero. Averaging the seven post-adoption coefficients yields declines of 4.5 percent for total arrests, 5.3 percent for violent arrests, and 4.7 percent for property arrests. These dynamic averages are smaller than the static estimates because the event study normalises on the year before adoption, whereas the static coefficient compares the post-adoption period with the pre-adoption period as a whole; the year before adoption sits about 0.04 log points below that average.

[Figure 4 about here]

Appendix Figure A3 reports the corresponding raw population-weighted arrest rates by event time. I present these unadjusted series as descriptive evidence; the causal estimates rely on the fixed effects, controls, and clean comparisons in equation (4).

6.4 Eligibility levers

Fifteen adoption stacks relaxed only the asset test, while seventeen also broadened income eligibility. I estimate both arms in a single fully interacted model, preserving the treated-comparison contrast within each stack while directly testing whether the estimates differ across the two policy packages.

Table 4 shows that asset-only expansions reduced total arrests by 10.3 percent, violent arrests by 9.3 percent, and property arrests by 11.0 percent. The corresponding estimates under asset-plus-income expansions are 6.4, 6.0, and 6.6 percent. The asset-only estimates are larger on all three indices, but none of the three differences approaches conventional significance, with p -values of 0.517, 0.597, and 0.461. Panel B is mixed rather than ordered: robbery, aggravated assault, and larceny are larger under the asset-only package, while murder, rape, burglary, and motor vehicle theft are larger under the broader one, and no component difference is significant. Because no contrast is statistically distinguishable, I read the lever split as descriptive of where the pooled effect appears to originate rather than as a ranking of the two eligibility screens.

Figure 5 shows the two arms over event time. The diagnostic evidence warrants caution in both arms, and the small number of stacks behind each contrast limits what the comparison can settle. Distinguishing the levers convincingly would require variation that separates them within a common calendar window, which the observed adoption record does not provide: states that removed the asset test alone adopted earlier on average than states that did both, so the lever contrast is partly a timing contrast.

[Table 4 about here]

[Figure 5 about here]

6.5 Heterogeneity by age, sex, and race

Table 5 decomposes the estimates by age, sex, and race. Each outcome measures a group's arrests relative to the agency's total covered population, so the coefficients describe proportional changes in that group's arrests rather than demographic-specific arrest risks.

Differences between groups come from fully interacted models estimated within the stacked sample.

The total-index decline is similar for adults ages 18–24 and 25–54, at 8.4 and 9.0 percent. The composition differs across the two crime categories. Violent arrests decline by 5.0 percent among young adults and 10.4 percent among prime-age adults, a statistically significant difference ($p = 0.002$). Property arrests decline by 9.6 and 9.2 percent, and that difference is not distinguishable from zero. Adults 55 and over serve as a check on the design rather than as a group the policy targets: they account for a small share of index arrests, they face different SNAP eligibility rules, and their estimates are smaller and indistinguishable from zero on all three indices, at 5.8, 4.1, and 4.4 percent. Figure 2 shows the three age series over the sample period.

The estimates are negative for both sexes across all three indices. Violent arrests decline by 8.3 percent among men and 6.4 percent among women, while property arrests decline by 7.4 and 11.1 percent. Neither difference by sex is significant at the 5 percent level.

The largest differences arise by race. Total, violent, and property arrests among White adults decline by 9.7, 8.9, and 10.4 percent. The corresponding estimates among Black adults are 2.5, 3.5, and 1.3 percent, and none is distinguishable from zero. The White–Black differences are statistically significant for all three indices, with p -values below 0.001 for the total and property indices and 0.002 for the violent index. Because the data provide neither demographic-specific participation responses nor links between SNAP receipt and arrests, these estimates identify reduced-form heterogeneity and do not establish why the responses differ.

[Table 5 about here]

6.6 Counterfactual policy magnitudes

I next translate the proportional estimates into policy quantities. Following the counterfactual approach in Wang (2026), I recover the no-BBCE path for each treated agency-year by removing the corresponding post-adoption event-time coefficient from the observed outcome. For a log outcome, this amounts to multiplying the observed level by $\exp(-\hat{\beta}_k)$. I calculate uncertainty using the delta method and the full covariance matrix of the post-adoption coefficients. The simulation runs only for indices whose joint lead test does not reject at the 5 percent level, which all three clear.

Figure 6 plots the resulting paths. Summed over the adoption year and six subsequent years, the estimates imply 18.3 million additional SNAP recipient-years across the thirty-

two adopting states, together with 211,931 fewer total-index arrests, 76,852 fewer violent arrests and 153,259 fewer property arrests among the treated agencies. These quantities represent 4.5, 5.2 and 4.7 percent of their respective counterfactual totals.⁶ Appendix A3 reports the estimates by calendar year with the corresponding confidence intervals.

Dividing one quantity by the other puts the magnitude on a scale the reader can judge: the estimates imply one averted index arrest for every 86 additional SNAP recipient-years, one averted property arrest for every 119, and one averted violent arrest for every 238. That is a large effect per participant relative to what the literature finds for benefit losses and benefit timing, and two features of the setting bear on whether it is plausible. Newly eligible households are a modest share of the caseload, so the implied within-group response is larger than the aggregate ratio suggests, and the estimate may also capture responses among households already receiving benefits whose expectations of continued eligibility changed. The design cannot separate these channels, and I therefore treat the ratio as an accounting summary of the reduced-form estimates rather than as a per-recipient treatment effect.

One additional assumption turns the scaling into an estimate. BBCE changes no criminal law, no policing practice, and no other transfer program, so if it moves arrests only by moving SNAP access, the participation coefficient in Table 2 is a first stage and the arrest coefficient in Table 3 a reduced form for the same instrument. Their ratio is then a just-identified instrumental-variables estimate of the elasticity of arrests with respect to SNAP participation. That elasticity is -1.15 for the total index, -1.05 for the violent index, and -1.22 for the property index, so a 10 percent increase in recipients corresponds to roughly an 11 percent fall in index arrests among the affected agencies. Because the two coefficients come from different panels, their covariance is not available in closed form; standard errors and percentile intervals come from a state block bootstrap of 299 draws that refits both regressions inside each draw.

Two things bound what the scaling adds. The first stage is not strong by the conventional rule, with an F statistic of 4.6, so the ratio is imprecise and its percentile interval on the total index runs from -5.30 to $+0.49$. And the exclusion restriction delivers an effect per additional recipient-year only if households newly made eligible and previously eligible households whose take-up changed respond alike. I therefore report the elasticity as the scale the reduced form implies under a stated assumption, not as a treatment effect for a well-defined complier group.

The two quantities also cover different geographic populations. The participation es-

⁶Because I estimate each logged index separately before converting its coefficients to levels, the simulated total-index count need not equal the sum of the separately simulated component counts.

timate covers the adopting states, whereas the arrest estimates cover sampled agencies representing a subset of the national population. I therefore report the arrest totals as sample-specific quantities and do not extrapolate them to unsampled jurisdictions.

[Figure 6 about here]

7 Robustness

I assess the main findings along five margins: the identifying assumptions, the comparison group, the estimator, the definition of treatment, and the sample and specification rules.

7.1 Pre-adoption trends and placebo estimates

Figure 4 provides an initial assessment of the parallel-trends assumption. The joint tests fail to reject the null hypothesis that the pre-adoption coefficients are jointly zero for the total and property indices, with p -values of 0.254 and 0.383. The violent leads are marginal at 0.076, and burglary rejects outright at 0.008. The evidence of differential pre-trends makes the burglary estimate difficult to interpret; accordingly, I treat it as uninformative and do not draw substantive conclusions from its negative static coefficient.

The lead tests are not conclusive. The earliest coefficients on all three indices are positive and decline toward zero before adoption, at roughly 0.08 log points four years out. Although the joint tests do not reject, such tests have limited power against gradual differential trends (Roth, 2022): with three lead coefficients and 44 clusters, the test fails to reject a drift of about 2.6 percent a year, which is a third the size of the estimated effect. Two further pieces of evidence bear on the question. The post-adoption path is not monotone. It falls sharply, holds for three years, and then reverts toward zero, a shape a smooth differential trend cannot produce. And an in-time placebo can be constructed directly.

Panel A of Table 8 moves each adoption date six years earlier and excludes every genuinely post-adoption observation, so the placebo window ends the year before true adoption. The placebo estimates are +2.3 percent for the total index, −7.8 percent for the violent index, and +6.5 percent for the property index, and none is statistically distinguishable from zero. The violent point estimate is not trivial, so the placebo is supportive rather than decisive. More importantly, the placebo produces neither the broad decline across all three indices nor the offense composition observed after actual adoption.

[Figure 8 about here]

7.2 Selection into adoption

States chose whether and when to implement BBCE, so a further concern is that changing crime conditions influenced policy timing. Table A3 regresses a state's adoption year on its 1996 to 1999 average characteristics, across the states adopting within the sample period, and repeats the regression weighting by population, adding Census region fixed effects, excluding California and New York, and excluding Southern states. No characteristic predicts timing in the baseline column, the joint test does not reject in any of the five, and pre-period characteristics explain 15 percent of the variation in timing in the baseline and no more than 24 percent in any column. The one coefficient reaching conventional significance is the minimum wage in the population-weighted column, which reflects the large early-adopting states on the West Coast and in the Northeast and disappears once region fixed effects enter. This evidence weighs against BBCE being adopted in response to the state conditions that also bear on arrests.

[Table A3 about here]

7.3 Differential linear trends

Table 7 asks whether the estimates survive when adopting states are allowed to trend differently from comparison states. Column (2) adds a linear trend in event time interacted with treatment status, so the coefficient on BBCE measures a level shift at adoption net of any drift the treated group shares. Column (3) adds a separate linear trend for every stack-by-state cell. Because the estimated effect varies over the post-adoption window, linear trends fitted over the full window absorb part of the effect itself (Wolfers, 2006), so both columns understate a genuine dynamic response. I therefore retain the baseline and treat the trend specifications as a bound on how much of the estimate a smooth differential trend could explain.

The total and property indices survive the treatment-group trend nearly intact, at 7.6 and 9.3 log points, and the treated-group slope is indistinguishable from zero for both, at -0.002 and $+0.000$ per year. Rape and larceny also retain their estimates. The violent index does not. Its treated-group slope is -0.011 per year and once that slope enters, the level shift falls to 1.8 log points and is far from significant. The sensitivity is concentrated in aggravated assault, whose slope is -0.014 per year and whose level shift falls from 9.4 to 2.0 log points. Adopting states were therefore already recording relatively fewer aggravated assault arrests in the years before adoption, and the post-adoption path is consistent with that drift continuing as well as with a break at adoption.

Column (3) is more demanding still, fitting 32 separate stack-by-state slopes through a window in which the effect itself moves. Total and property remain significant at 4.0 and 4.4 log points, rape and burglary retain theirs, and the remaining estimates shrink toward zero. There is no column that adds both trend sets together, because the two are not separately identified: within a stack the treated state occupies its own stack-by-state cell, so the trend fitted to that cell is the treatment-group trend, and estimating both returns column (3) exactly. Appendix Table [A2](#) reports the component offenses.

I read the table as follows. The property results, and the total index that they dominate, are robust to differential linear trends. The violent results are not, and the paper's claims about violent arrests, and about aggravated assault in particular, should be read as conditional on the parallel-trends evidence from the leads and the placebo rather than as robust to trend controls.

[Table 7 about here]

7.4 Comparison states

The baseline comparison pool includes states that never adopted BBCE and states that remained untreated throughout the focal cohort's event window. Table [6](#) restricts the comparison group to states that never adopted during the sample period. The average post-adoption declines are 9.7 percent for the total index, 11.8 percent for the violent index, and 9.1 percent for the property index, each larger than the corresponding baseline estimate rather than smaller, and the pre-adoption coefficients become flatter, with joint-test p -values of 0.450, 0.493, and 0.134. Both panels rest on the seventeen cohorts for which a never-treated comparison group of sufficient size exists. The results therefore do not depend on using later adopters as comparisons.

[Table 6 about here]

[Figure 7 about here]

7.5 Estimator

The stacked design addresses the negative-weighting problem directly: each stack contains one treated cohort and comparison agencies that are either never treated or adopt more than six years later, so no already-treated unit ever serves as a control. Appendix Table [A5](#) shows what that restriction is worth. On a balanced agency-year panel, the two-way fixed effects coefficient on the total index is +0.006, essentially zero and of the wrong

sign. Decomposing it in the manner of Goodman-Bacon (2021), the clean treated-versus-never-treated comparisons average -0.053 and carry 49 percent of the weight, while the two comparison types that use another adopting state as the control average $+0.059$ and $+0.067$ and carry the remaining 51 percent. The contaminated comparisons are what the stacking removes.

Table 9 reports the imputation estimator of Borusyak et al. (2024), applied to the same stacks the baseline uses so that the estimator is the only thing that changes. Fixed effects and controls are fitted on untreated cells alone, the counterfactual is imputed for each treated post-adoption cell, and standard errors come from a state block bootstrap that refits the imputation in every draw. The imputation estimates are larger than the baseline, at 11.7, 9.3, and 12.7 log points, and remain significant at the 10 percent level. Because the imputation weights every treated cell by population rather than through the stacked comparison structure, it loads more heavily on the middle post-adoption years, where the estimated effect is deepest. The baseline is therefore the more conservative of the two.

[Table 9 about here]

7.6 Treatment definition, sample, and functional form

Appendix Table A1 varies the remaining design choices.

Panel A varies the minimum covered population. Removing the floor, or raising it to 25,000 or 50,000, leaves the total-index estimate between 8.6 and 9.7 log points. The floor exists for the component table rather than the indices: without it, 70 percent of murder agency-years and 50 percent of rape agency-years are zeros that the log drops, so each component would be estimated on a different self-selected roster.

Panel B varies the definition of treatment. Adoption month ranges from January to December and the adoption year is on average only 55 percent treated, so the baseline coding of $D = 1$ from the adoption year understates the effect. Coding that year as the fraction in force gives 9.1, 8.6, and 9.5 log points, and dropping it gives 9.7, 9.0, and 10.2. Excluding Louisiana, whose 2014 reversal falls inside its window, changes nothing.

Panel C varies functional form. Poisson pseudo-maximum likelihood on counts with log covered population as an offset, which retains the zero cells the log drops, gives 6.9, 6.9, and 6.0 log points, with the violent estimate losing significance. The inverse hyperbolic sine reproduces the log almost exactly.

Panel D restores the five-lead window of an earlier draft. That window requires five years of pre-adoption data and so excludes the four states adopting in 2000, leaving 24

stacks and 602 agencies; it returns 12.5, 12.9, and 12.6 log points. The current design is the more conservative one, and it uses every adopter the data can support.

7.7 Enforcement capacity

Arrests are produced jointly by offenders and police, so a decline could reflect fewer officers rather than less offending. Table A4 addresses this directly. Panel A replaces the arrest rate with police staffing: sworn officers per 100,000 residents fall by 3.8 log points in adopting states and total employees by 3.6, with no pre-trend and no change in the civilian share of the force. Panel B adds staffing to the arrest regression. Conditioning on sworn officers moves the total-index estimate by 0.003 log points and the property estimate by 0.002. The measured elasticity of arrests with respect to sworn officers is 0.06 to 0.10, so a 3.8 percent staffing decline accounts for roughly 0.3 log points of an 8.7-point effect. Staffing is itself an outcome of state fiscal conditions and moves with adoption, so Panel B is a bound on the enforcement channel rather than a preferred specification; the bound is small.

[Table A4 about here]

7.8 Adoption timing and the Great Recession

Adoption was concentrated around the Great Recession and formed part of a broader modernization of SNAP administration, so BBCE coincided with other policy and economic changes. Every specification controls for six contemporaneous SNAP policies, including simplified reporting, vehicle exclusions, EBT issuance, and online applications, as well as state economic and income-support conditions, and the stack-by-year fixed effects absorb shocks shared by treated and comparison agencies within a cohort and calendar year. Twenty-one of the thirty-two adoption events occurred between 2008 and 2010, raising the further possibility that the estimates reflect the conditions surrounding the Great Recession. Panel C of Table 8 estimates the effects separately for the eleven cohorts adopting between 2000 and 2007 and the twenty-one later cohorts.

The two eras differ by outcome. Violent arrests fall by essentially the same amount in both, 8.2 and 7.8 log points, with a difference of 0.4 log points. Property arrests fall by 1.9 log points among the earlier cohorts and 14.1 among the later ones, and the total index follows property. Neither difference is significant at the 5 percent level, and the property difference reaches $p = 0.096$.

This comparison has a limited interpretation because adoption era is nearly collinear with the macroeconomic cycle and with policy scope, so a difference between the panels has more than one reading. Its narrower contribution is to show that the violent result is present in both eras while the property result is concentrated among states that expanded eligibility as the downturn arrived. That the property effect is larger where and when need was greater is what a resource mechanism predicts, and it is the opposite of what recession contamination alone would produce, since worse labour markets raise rather than lower property offending.

[Table 8 about here]

8 Discussion

BBCE changed access to SNAP without changing the federal benefit formula. The results therefore identify eligibility design as a consequential margin of the program. Adoption increased participation and reduced adult arrests across the total, violent, and property indices. The similar proportional declines in violent and property arrests suggest that no single mechanism explains the full response.

The property results are consistent with a resource channel. Larceny accounts for most of the decline within the property index. Greater access to food assistance may reduce immediate material shortfalls, free cash for other necessities, and help households preserve transportation or savings that support employment. This interpretation accords with evidence that property crime responds to the timing and delivery of transfers (Foley, 2011; Carr and Packham, 2019; Watson et al., 2020; Wright et al., 2017). The present findings extend that relationship to the rules governing program entry.

The violent results require a broader explanation. Rape and aggravated assault also decline substantially. Financial strain can increase stress and interpersonal conflict even when material gain is not the object of an offense, and more stable access to food assistance may reduce those pressures. This interpretation is consistent with evidence linking the SNAP benefit cycle to domestic violence (Carr and Packham, 2021) and with the lower rates of relationship violence observed in states with broader BBCE policies (Austin et al., 2023). The UCR does not identify the relationship between an arrestee and a victim, however, so the offense pattern supports but does not isolate a household-stability channel.

The demographic results provide additional, though still indirect, evidence. Violent arrests decline more among adults ages 25–54 than among adults ages 18–24, despite the higher individual propensity for arrest among younger adults (Hirschi and Gottfredson,

1983). This ordering does not fit a mechanism operating solely through the offending incentives of marginal young adults and is consistent with a policy affecting household resources and stability. The estimates do not differ statistically by sex. Arrests decline among both White and Black adults, but the reductions are larger among White adults. Differences in eligibility, take-up, adoption geography, or exposure to the criminal justice system could produce this pattern. Without demographic-specific participation estimates or linked household records, the data cannot distinguish among these explanations.

The policy-lever estimates provide a final clue, and a weak one. Asset-only expansions carry larger point estimates on all three indices than asset-plus-income expansions, and most component offenses follow the same ordering. No contrast approaches conventional significance, however, and the two arms differ in adoption timing as well as in reach, so the lever split is partly a timing split. Evidence that asset and income rules reach different populations and generate different participation responses (Han and Wu, 2026) makes the ordering worth reporting, but the decomposition indicates where the pooled estimates appear to originate rather than ranking the two screens.

These findings broaden how eligibility rules should be evaluated. Asset tests, income limits, and verification requirements are commonly assessed according to their effects on targeting, enrollment, and program costs. Research on administrative burden shows that these rules also determine who successfully receives assistance (Currie, 2006; Deshpande and Li, 2019; Finkelstein and Notowidigdo, 2019; Homonoff and Somerville, 2021). The results here indicate that their consequences can extend beyond the program and into criminal justice contact. The relevant policy accounting therefore includes not only caseload and expenditure effects but also outcomes borne by households and communities.

The counterfactual exercise illustrates the scale of this additional margin rather than providing a cost-benefit calculation. It implies approximately 17 million additional SNAP recipient-years across the adopting states and 212,000 fewer total-index arrests across the sampled agencies during the adoption year and six subsequent years, or one averted index arrest for every 81 additional recipient-years. That ratio is large relative to what the literature finds for benefit losses and benefit timing, and Section 6.6 sets out why: newly eligible households are a modest share of the caseload, so the implied within-group response is larger still, and the estimate may also capture responses among households whose expectations of continued eligibility changed. The participation estimate is statewide, whereas the arrest estimate applies to the treated agencies in the analysis, so the two should be read as an accounting summary rather than extrapolated into a national total.

The evidence also does not prescribe one form of BBCE over another. The results identify expanded access as the relevant policy margin but do not determine which screen produces the greatest downstream response per newly enrolled household.

8.1 Limitations

The estimates are reduced-form effects on treated jurisdictions rather than effects on individual SNAP participants. Section 6.6 scales them by the first stage under an exclusion restriction and reports the implied elasticity, but the first stage is not strong enough for that ratio to be precise, and the scaling still cannot separate newly eligible participation from greater take-up among previously eligible households. Recovering an effect for a well-defined complier group, or explaining the demographic differences, would require data linking SNAP receipt to arrests at the household level.

The sample also limits external validity. Agencies must report twelve months in every year of a stack window to enter, so departments that file intermittently are absent and small jurisdictions are underrepresented. Removing the covered-population floor entirely leaves the estimates unchanged, so it is the reporting requirement rather than agency size that defines the sample. The overlap between policy scope and adoption timing makes the lever comparisons suggestive rather than definitive. Counterfactual arrest totals apply only to sampled agencies and inherit the identifying assumptions of the underlying estimates.

9 Conclusion

Means-tested programs are shaped by both the benefits they provide and the rules determining who may receive them. Existing research on SNAP and crime has concentrated primarily on benefit amounts, timing, delivery, and withdrawal. By exploiting state variation in BBCE adoption, this paper examines the eligibility margin.

The findings indicate that expanding SNAP eligibility increased participation and reduced adult arrests for violent and property offenses. The precise mechanisms remain unresolved, but the central implication is clear. Eligibility rules shape more than enrollment and program expenditures. Evaluating them solely by whom they exclude and what they save overlooks consequences that arise beyond the program itself.

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Tables and Figures

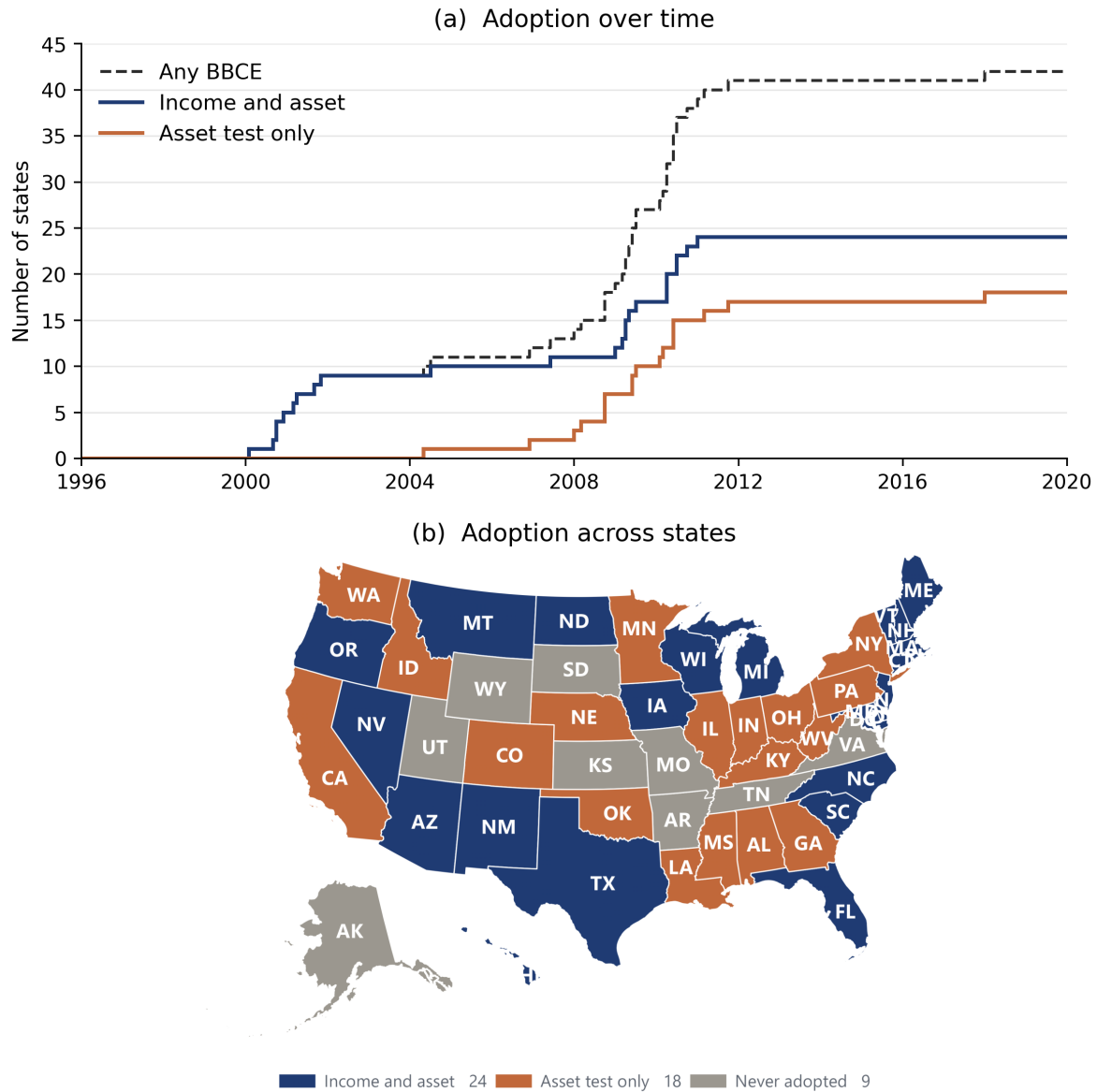


Figure 1: BBCE variation over time and across states, 1996–2019

Notes: Panel (a) plots the cumulative number of states with BBCE in force in a given month, separately by lever, with the combined count shown as Any BBCE. Panel (b) maps the same classification. The estimating sample ends in 2016 and uses the 32 cohorts whose event windows fit inside it.

Table 1: Summary statistics

Variable	All agencies		Adopting states		Comparison states	
	Mean	SD	Mean	SD	Mean	SD
<i>Arrests per 100,000 covered residents, adults</i>						
Total index	608.4	392.0	605.5	367.4	619.2	469.8
Violent index	197.5	163.6	209.7	163.6	153.6	156.0
Murder	4.6	7.9	4.6	8.3	4.2	6.6
Rape	7.0	8.5	7.2	8.7	6.5	7.7
Robbery	34.2	34.9	36.8	34.9	25.1	33.5
Aggravated assault	151.7	134.5	161.1	135.2	117.7	126.5
Property index	410.9	308.2	395.8	286.7	465.6	370.3
Burglary	78.2	59.9	83.1	61.4	60.5	50.3
Larceny	302.1	275.0	280.6	254.5	379.6	327.4
Motor vehicle theft	30.6	34.9	32.1	34.0	25.5	37.4
<i>Agency and state characteristics</i>						
Agency covered population	390,613	747,586	449,365	826,750	179,082	230,203
State unemployment rate	6.16	2.11	6.39	2.13	5.33	1.81
State poverty rate	13.05	2.96	13.36	2.90	11.91	2.90
Log minimum wage	1.83	0.20	1.84	0.20	1.79	0.17
Log maximum TANF benefit	6.07	0.43	6.16	0.41	5.74	0.35
State EITC rate	0.04	0.09	0.04	0.09	0.05	0.08
Democratic governor	0.44	0.50	0.43	0.49	0.50	0.50
Log state population	16.20	0.98	16.42	0.96	15.41	0.57
Simplified reporting	0.55	0.50	0.49	0.50	0.77	0.42
Combined application project	0.22	0.41	0.23	0.42	0.17	0.38
Vehicle exclusion	0.74	0.44	0.73	0.45	0.80	0.40
EBT issuance	0.85	0.36	0.84	0.37	0.89	0.31
Online application	0.41	0.49	0.39	0.49	0.48	0.50
Short recertification	0.94	0.24	0.95	0.23	0.92	0.27

Notes: Population-weighted means and standard deviations over the estimation sample, with each agency-year counted once. Adopting states are the 32 that adopt BBCE inside the sample window; comparison states are the 12 that never do. The sample is 2,784 agencies in 44 states, 32 stacks, 255,640 stack-agency-year observations.

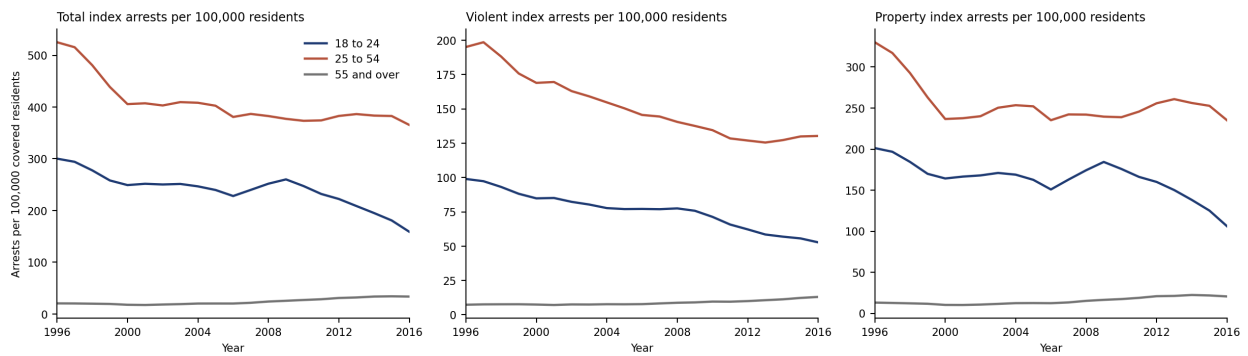


Figure 2: Adult arrest rates by age group, balanced agency roster

Notes: Arrests per 100,000 covered residents for adults 18 to 24, 25 to 54, and 55 and over, for the three composite indices. The roster is the 782 agencies reporting twelve months in every year from 1996 to 2016 with mean covered population of at least 10,000, covering 60.5 million residents in 1996 and 73.3 million in 2016. A balanced roster is necessary here: an agency-year enters the estimation sample only if it falls inside some stack's window, so that sample's composition varies from 1,382 agencies in 1996 to 2,784 in 2006 and 781 in 2016, and movement at the ends of a series drawn from it would be composition rather than trend.

Table 2: BBCE and SNAP participation

Outcome	BBCE in force	Event-study mean	Pre-trend p
Log SNAP recipients	0.075** (0.035)	0.092** (0.036)	0.391
Log SNAP household caseloads	0.087*** (0.027)	0.087*** (0.024)	0.854
Stacks	32	32	
State-years	5,904	5,904	

Notes: State-level analogue of the agency design: one stack per adopting state, the same comparison rule and the same window, with state-year observations weighted by state population and standard errors clustered on state. The event-study mean averages the seven post-adoption coefficients and the pre-trend p -value is the joint test of the three leads. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

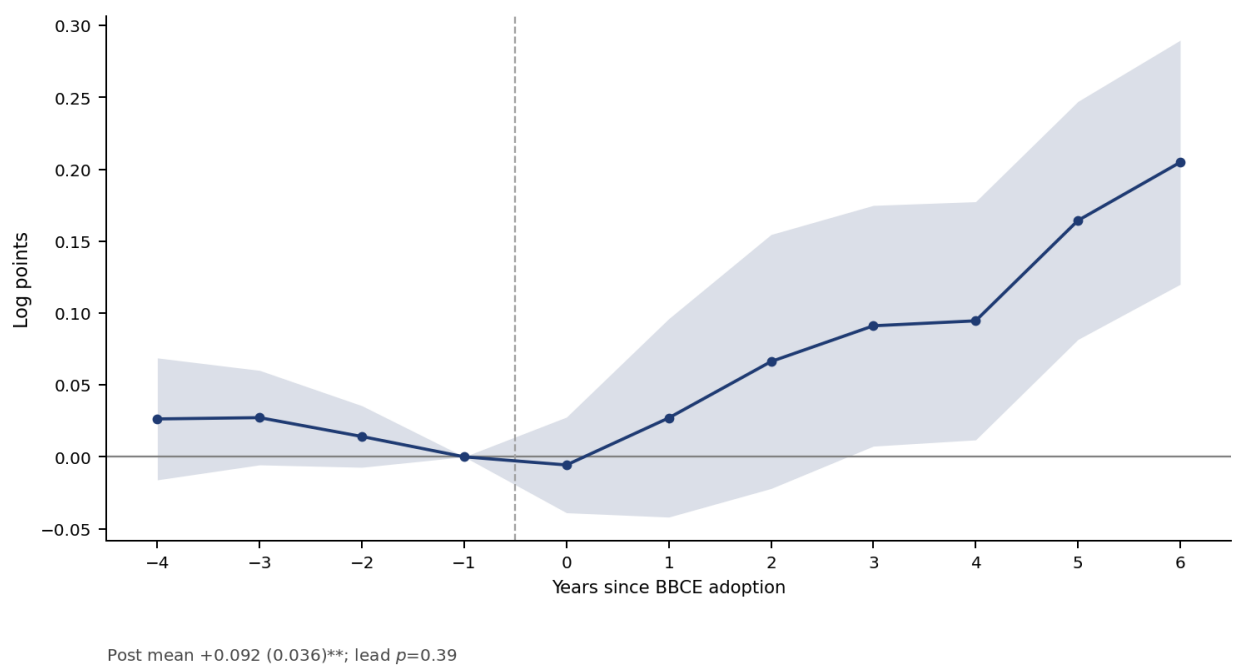


Figure 3: BBCE and SNAP participation, event study

Notes: Coefficients on leads $k = -5$ through -2 and lags $k = 0$ through 6 , with $k = -1$ normalised to zero, on log SNAP recipients per capita. Shaded band is the 95 percent confidence interval.

Table 3: BBCE and arrests of adults

Outcome	BBCE in force	Event-study mean	Pre-trend p
Total index	−0.087** (0.043)	−0.046* (0.026)	0.254
<i>Violent index and its components</i>			
Violent index	−0.079** (0.036)	−0.054** (0.026)	0.076
Murder	−0.031 (0.040)	−0.029 (0.040)	0.358
Rape	−0.125** (0.059)	−0.111** (0.050)	0.844
Robbery	−0.031 (0.024)	−0.024 (0.020)	0.268
Aggravated assault	−0.094* (0.049)	−0.064* (0.038)	0.036
<i>Property index and its components</i>			
Property index	−0.092** (0.046)	−0.048* (0.026)	0.383
Burglary	−0.046 (0.034)	0.010 (0.028)	0.008
Larceny	−0.103** (0.052)	−0.072** (0.032)	0.461
Motor vehicle theft	−0.118 (0.073)	−0.042 (0.077)	0.304

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with stack $\times$ agency and stack $\times$ year fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. The event-study mean averages the seven post-adoption coefficients from equation (4) and the pre-trend p -value is the joint test of the three leads. The three indices are the primary outcomes; the component offenses describe the composition of an effect the indices establish. The sample is 2,784 agencies in 44 states, 32 stacks, 255,640 stack-agency-year observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

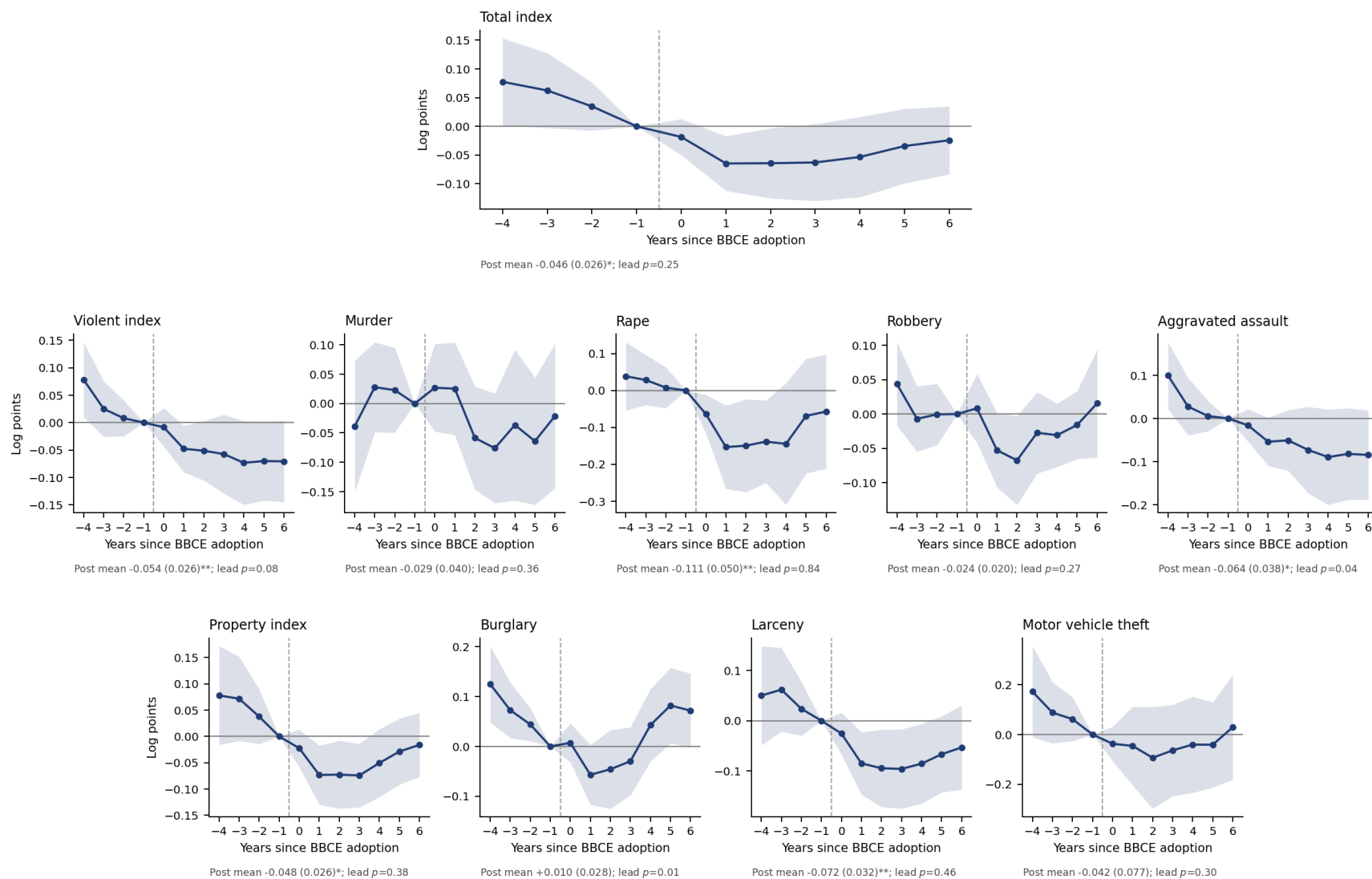


Figure 4: Event-study estimates, arrests of adults

Notes: Coefficients from equation (4), weighted by agency covered population. Shaded bands are 95 percent confidence intervals. Panel notes report the mean of the seven post-adoption coefficients and the joint test that the four leads are zero.

Table 4: Which eligibility lever?

Outcome	Asset test only	Income and asset	Difference (<i>p</i>)
<i>Panel A. Composite indices</i>			
Total index	−0.109* (0.064)	−0.066 (0.041)	0.042 (0.517)
Violent index	−0.098* (0.054)	−0.062 (0.045)	0.036 (0.597)
Property index	−0.117* (0.065)	−0.068 (0.045)	0.049 (0.461)
<i>Panel B. Component offenses</i>			
Murder	−0.013 (0.047)	−0.047 (0.057)	−0.033 (0.632)
Rape	−0.104 (0.095)	−0.145** (0.060)	−0.041 (0.690)
Robbery	−0.061* (0.033)	−0.003 (0.027)	0.058 (0.118)
Aggravated assault	−0.113* (0.067)	−0.077 (0.060)	0.036 (0.660)
Burglary	−0.021 (0.045)	−0.068 (0.045)	−0.047 (0.426)
Larceny	−0.146** (0.073)	−0.063 (0.048)	0.083 (0.248)
Motor vehicle theft	−0.089 (0.104)	−0.144 (0.088)	−0.054 (0.663)
Stacks	15	17	

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with $\text{stack} \times \text{agency}$ and $\text{stack} \times \text{year}$ fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. The treatment indicator is interacted with the policy package a state adopted, so both arms come from one regression and the difference is a linear contrast within it. Asset-only states removed the asset test while leaving the gross income limit at the federal 130 percent of the poverty line; asset-plus-income states did both. The sample is 2,784 agencies in 44 states, 32 stacks, 255,640 stack-agency-year observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

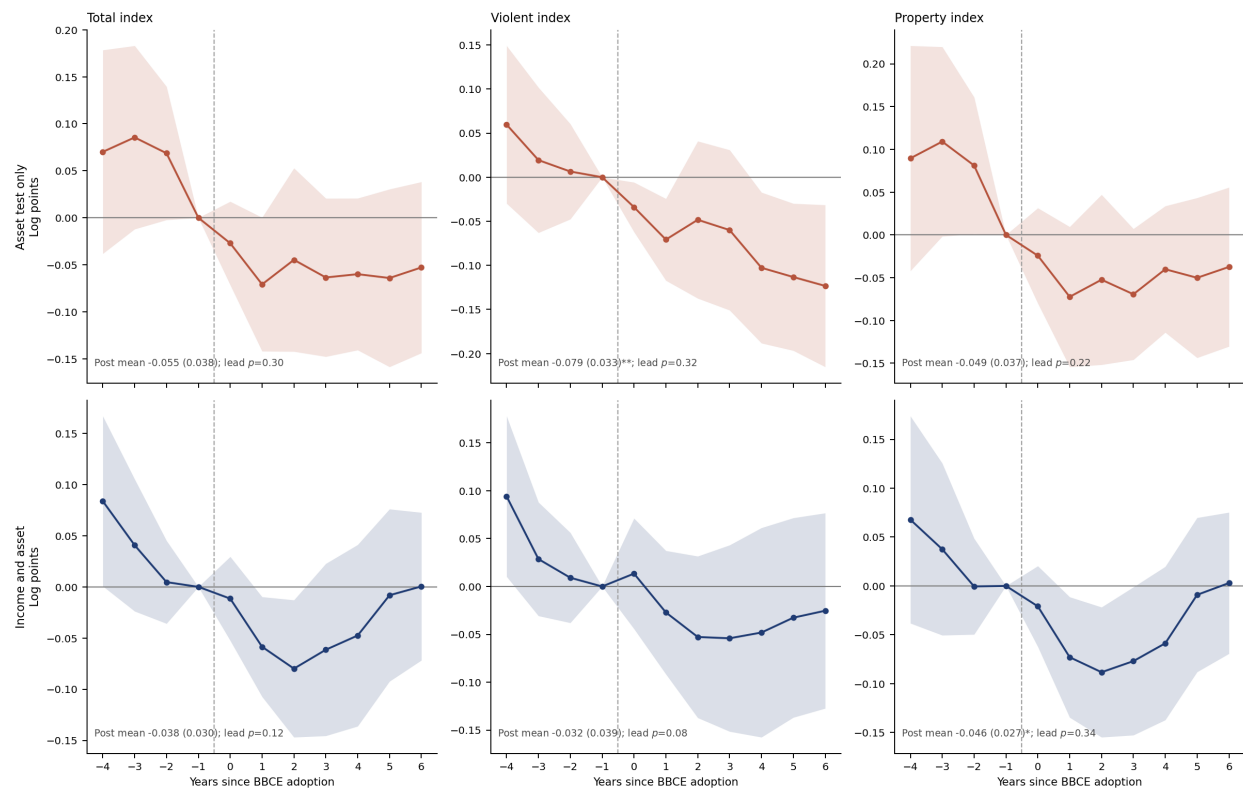


Figure 5: Event-study estimates by eligibility lever

Notes: Leads and lags are interacted with the lever arm, so both rows come from a single regression. Top row is the eleven stacks that also raised the gross income threshold; bottom row is the thirteen that eliminated the asset test alone.

Table 5: Age, sex, and race of arrestees

Group	Total index	Violent index	Property index
<i>Panel A. Age</i>			
18 to 24	−0.088* (0.049)	−0.051 (0.035)	−0.101* (0.055)
25 to 54	−0.094** (0.041)	−0.110*** (0.037)	−0.096** (0.043)
55 and over	−0.060 (0.054)	−0.042 (0.056)	−0.045 (0.060)
<i>Panel B. Sex</i>			
Male	−0.076** (0.039)	−0.087** (0.036)	−0.076* (0.041)
Female	−0.102** (0.052)	−0.066 (0.046)	−0.117** (0.053)
<i>Panel C. Race</i>			
White	−0.102** (0.041)	−0.093** (0.039)	−0.110** (0.043)
Black	−0.026 (0.041)	−0.036 (0.032)	−0.014 (0.047)

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with stack $\times$ agency and stack $\times$ year fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. Each cell is the logged arrest rate for the named group, using that group's arrests over the agency's whole covered population, so the estimates are comparable across groups and are not affected by group population shares. The sample is 2,784 agencies in 44 states, 32 stacks, 255,640 stack-agency-year observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

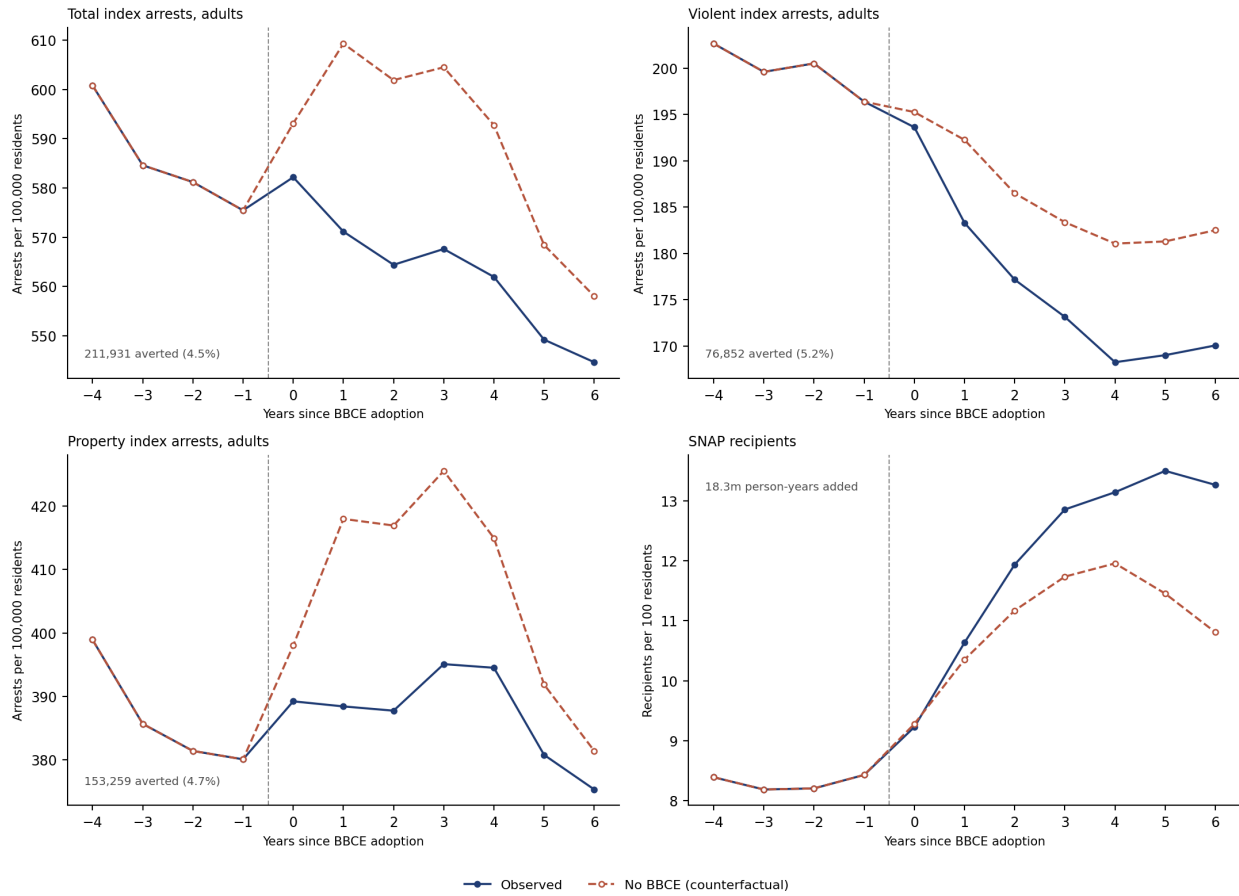


Figure 6: Observed and counterfactual paths

Notes: The counterfactual removes the estimated event-time coefficient from each treated agency-year, so the two series coincide before adoption by construction. The panels report adult arrests per 100,000 covered residents across the treated agencies for each index whose joint lead test does not reject at the 5 percent level, together with SNAP recipients per 100 residents in the same states. All three indices clear that bar, though the violent leads are marginal at $p = 0.076$. In-panel figures give the total averted over $k = 0$ to 6 and its share of the counterfactual level.

Table 6: Comparison group

Outcome	Event-study mean	Standard error	Pre-trend p
<i>Panel A. Never-treated and later-treated comparison states</i>			
Total index	−0.089**	(0.041)	0.505
Violent index	−0.124***	(0.043)	0.438
Property index	−0.081*	(0.043)	0.138
<i>Panel B. Never-treated comparison states only</i>			
Total index	−0.102**	(0.043)	0.450
Violent index	−0.126***	(0.046)	0.493
Property index	−0.096**	(0.044)	0.134

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with stack $\times$ agency and stack $\times$ year fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. Panel A repeats the baseline comparison rule, in which a state enters a stack as a comparison if it never adopts or adopts more than six years after the focal state. Panel B keeps only states that never adopt within the sample period, which removes any possibility that a later adopter's own anticipation contaminates the comparison. Both panels use 17 stacks, the cohorts for which a never-treated comparison group of sufficient size exists. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

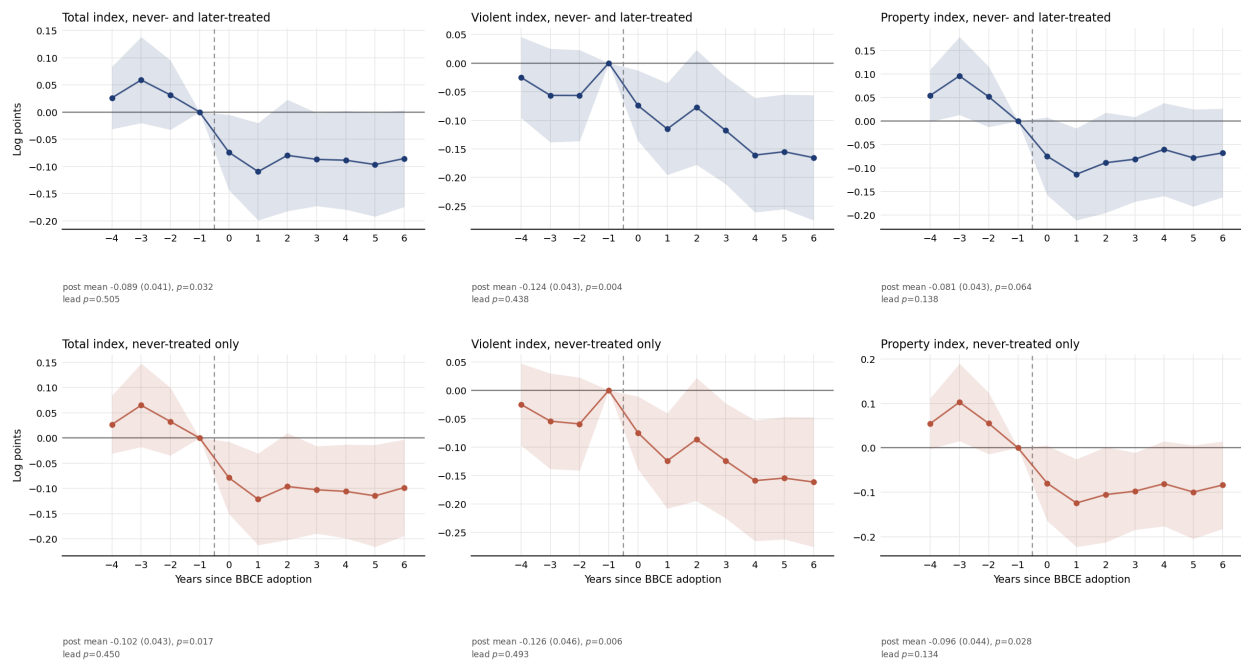


Figure 7: Event-study estimates under both comparison groups

Notes: Top row uses the baseline comparison group of never- and later-treated states; bottom row restricts to the nine never-treated states.

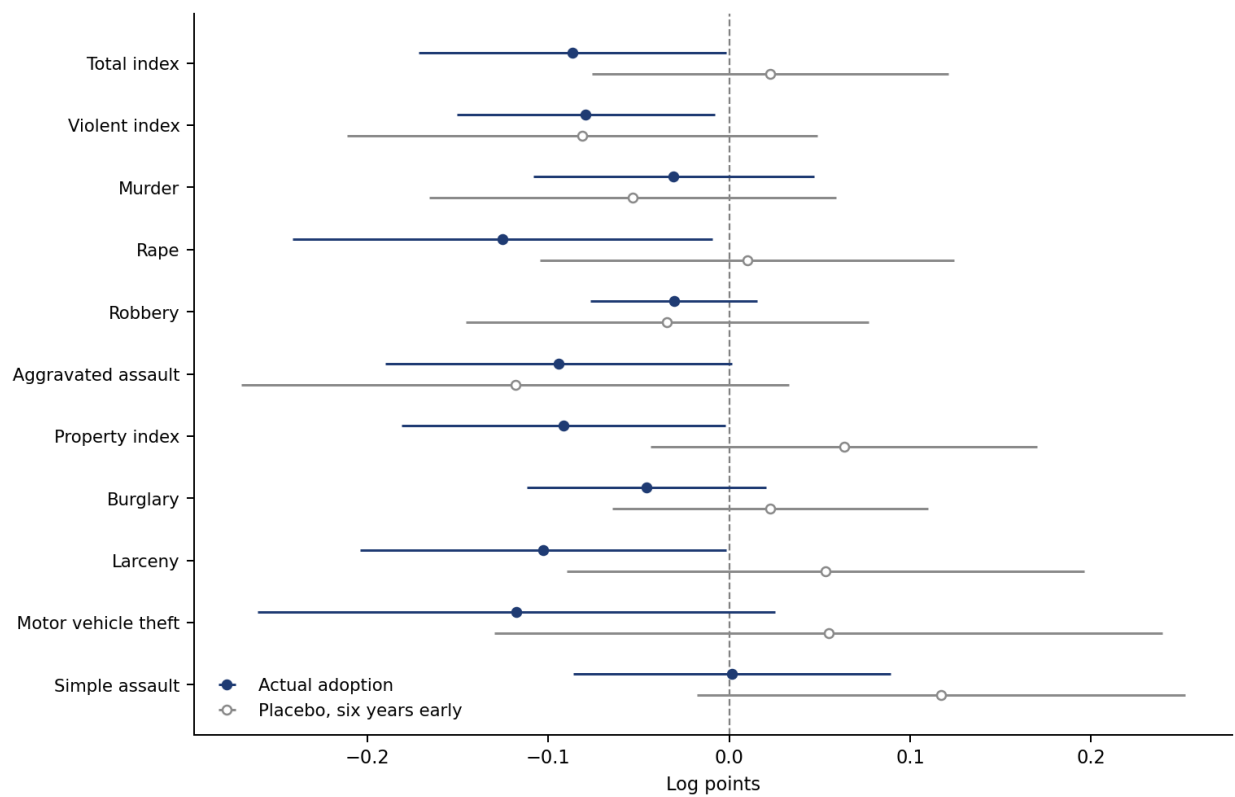


Figure 8: In-time placebo: estimates at true and falsified adoption dates

Notes: Navy markers reproduce the baseline estimates; grey markers move every state's adoption date back by six years and truncate the sample so only genuinely pre-adoption data enter.

Table 7: Linear trends

Outcome	Baseline	Treatment-group trend	Stack $\times$ state trends
Total index	−0.087** (0.043)	−0.076*** (0.028)	−0.040** (0.018)
Violent index	−0.079** (0.036)	−0.018 (0.029)	−0.014 (0.024)
Property index	−0.092** (0.046)	−0.093** (0.036)	−0.044* (0.024)
Stack $\times$ agency FE	Yes	Yes	Yes
Stack $\times$ year FE	Yes	Yes	Yes
State-year controls	Yes	Yes	Yes
Treatment-group linear trend	No	Yes	No
Stack $\times$ state linear trends	No	No	Yes
Population weights	Yes	Yes	Yes
State clusters	Yes	Yes	Yes

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with stack $\times$ agency and stack $\times$ year fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. Column (2) adds a linear trend in event time interacted with treatment status, so the treated group is allowed its own slope; column (3) adds a separate linear time trend for every stack-by-state cell. Component offenses appear in Appendix Table A2. The sample is 2,784 agencies in 44 states, 32 stacks, 255,640 stack-agency-year observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Placebo, weighting, and adoption era

Outcome	(1)	(2)	(3)
<i>Panel A. In-time placebo, adoption assigned six years early</i>			
Total index	0.023	(0.050)	−0.087**
Violent index	−0.081	(0.066)	−0.079**
Property index	0.063	(0.055)	−0.092**
<i>Panel B. Unweighted</i>			
Total index	−0.063**	(0.028)	−0.087**
Violent index	−0.061**	(0.030)	−0.079**
Property index	−0.064*	(0.035)	−0.092**
<i>Panel C. Adoption era</i>			
Total index	−0.040 (0.043)	−0.119* (0.063)	−0.079 (0.307)
Violent index	−0.082 (0.052)	−0.078 (0.056)	0.004 (0.961)
Property index	−0.019 (0.041)	−0.141** (0.063)	−0.122 (0.096)

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with stack $\times$ agency and stack $\times$ year fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. In Panel A column (1) is the placebo estimate and column (3) the actual estimate; adoption is moved back six years and every genuinely post-adoption observation is dropped, so the placebo window ends the year before true adoption and a correctly specified design returns zero. Seventeen stacks have enough pre-adoption data to enter. In Panel B column (1) drops the population weights and column (3) repeats the weighted baseline. In Panel C column (1) covers the eleven cohorts adopting between 2000 and 2007, column (2) the twenty-one adopting between 2008 and 2010, and column (3) their difference; adoption era is close to collinear with the macroeconomic cycle, so the split shows whether the pooled result is confined to recession-era adopters rather than identifying heterogeneity by economic conditions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Imputation estimator

Outcome	Stacked	Imputation	Percentile interval
Total index	−0.087** (0.043)	−0.117* (0.060)	[−0.204, 0.033]
Violent index	−0.079** (0.036)	−0.093* (0.048)	[−0.170, 0.007]
Property index	−0.092** (0.046)	−0.127* (0.066)	[−0.220, 0.039]

Notes: The imputation estimator of Borusyak et al. (2024) is applied to the same stacks the baseline uses. $\text{Stack} \times \text{agency}$ and $\text{stack} \times \text{year}$ effects and the controls are fitted on untreated cells only, meaning comparison agencies at every event time and treated agencies before adoption; the counterfactual is then imputed for each of the 13,559 treated post-adoption cells and the estimate is the population-weighted mean of the residuals. Standard errors and the percentile interval come from 200 state block bootstrap draws in which the imputation is refitted, so the first-step estimation error enters the variance. The sample is 2,784 agencies in 44 states, 32 stacks, 255,640 stack-agency-year observations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

A1 Additional robustness

Table A1: Sensitivity to the sample and specification rules

Specification	Total index	Violent index	Property index
<i>Panel A. Minimum covered population</i>			
Baseline, 10,000	−0.087** (0.043)	−0.079** (0.036)	−0.092** (0.046)
No floor	−0.086** (0.041)	−0.073** (0.035)	−0.091** (0.043)
25,000	−0.087* (0.050)	−0.080* (0.041)	−0.093* (0.051)
50,000	−0.097* (0.054)	−0.093* (0.052)	−0.102* (0.054)
<i>Panel B. Treatment definition</i>			
Fractional adoption year	−0.091** (0.045)	−0.086** (0.039)	−0.095** (0.047)
Dropping the adoption year	−0.097** (0.046)	−0.090** (0.040)	−0.102** (0.048)
Excluding Louisiana	−0.085** (0.044)	−0.075** (0.036)	−0.091** (0.046)
<i>Panel C. Functional form</i>			
Poisson with a population offset	−0.069** (0.030)	−0.069 (0.043)	−0.060* (0.033)
Inverse hyperbolic sine	−0.086** (0.043)	−0.086** (0.038)	−0.092** (0.045)
<i>Panel D. Estimation window</i>			
Five leads, 24 stacks	−0.125** (0.055)	−0.129*** (0.048)	−0.126** (0.057)

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with stack $\times$ agency and stack $\times$ year fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. Panel A varies the minimum covered population an agency must average over the stack window. Panel B codes the adoption year as the fraction of it the policy was in force, then drops it, then excludes Louisiana, whose 2014 reversal falls inside its window. Panel C replaces the log rate with Poisson pseudo-maximum likelihood on counts with log covered population as an offset, which retains the zero cells the log drops, and with the inverse hyperbolic sine. Panel D restores the five-lead window of an earlier draft, which excludes the four states adopting in 2000 and holds only 24 stacks and 602 agencies. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Linear trends, component offenses

Outcome	Baseline	Treatment-group trend	Stack $\times$ state trends
Murder	−0.031 (0.040)	0.010 (0.040)	0.036 (0.043)
Rape	−0.125** (0.059)	−0.149** (0.060)	−0.104** (0.051)
Robbery	−0.031 (0.024)	−0.044 (0.043)	−0.027 (0.040)
Aggravated assault	−0.094* (0.049)	−0.020 (0.035)	−0.024 (0.031)
Burglary	−0.046 (0.034)	−0.107*** (0.036)	−0.068** (0.030)
Larceny	−0.103** (0.052)	−0.083** (0.038)	−0.028 (0.023)
Motor vehicle theft	−0.118 (0.073)	−0.118 (0.084)	−0.060 (0.080)

Notes: Component-offence version of Table 7; specifications and sample are identical. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Determinants of BBCE adoption timing

	Baseline	Weighted	Region FE	Excl. CA, NY	Excl. South
Unemployment rate	−0.214 (1.043)	1.744 (1.624)	−0.024 (1.384)	−0.344 (1.064)	1.393 (1.488)
Poverty rate	0.480 (0.383)	0.260 (0.394)	0.581 (0.458)	0.410 (0.423)	0.061 (0.397)
Log minimum wage	−33.320 (25.602)	−63.671*** (23.766)	−36.539 (29.823)	−33.796 (25.799)	−47.660 (31.008)
Log TANF benefit	2.009 (2.429)	3.902 (3.610)	−2.284 (3.282)	1.512 (2.591)	−1.981 (3.813)
State EITC rate	−4.321 (9.414)	−16.306* (9.094)	−2.082 (8.800)	−4.658 (10.809)	10.973 (10.256)
Democratic governor	1.260 (1.646)	2.649 (1.760)	3.844* (2.294)	1.135 (1.676)	3.248 (2.499)
Log population	−0.487 (0.712)	−0.957 (0.846)	−0.209 (0.717)	−0.721 (0.992)	−0.323 (0.938)
Log SNAP per capita	−1.180 (3.593)	−3.648 (5.914)	−0.833 (5.670)	−0.761 (3.829)	−3.231 (4.732)
Joint test of all characteristics, p	0.583	0.159	0.186	0.494	0.678
R^2	0.154	0.240	0.243	0.159	0.228
Adopting states	38	38	38	36	25

Notes: The dependent variable is the calendar year in which a state adopted BBCE, regressed on that state's 1996 to 1999 average characteristics across the states that adopt within the sample period. Column (1) is the baseline; column (2) weights by population; column (3) adds Census region fixed effects; column (4) excludes California and New York; column (5) excludes Southern states. Robust standard errors in parentheses. No characteristic predicts timing in the baseline and the joint test never rejects, which supports treating adoption timing as unrelated to pre-period state conditions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Enforcement capacity and the arrest measure

	(1)	(2)	(3)
<i>Panel A. Does BBCE change enforcement capacity?</i>			
Log sworn officers per 100,000	−0.038***	(0.011)	0.266
Log total employees per 100,000	−0.036**	(0.015)	0.759
Civilian share of employees	0.002	(0.004)	0.591
<i>Panel B. Arrest estimates conditioning on staffing</i>			
Total index	−0.098**	−0.095**	−0.096**
	(0.040)	(0.040)	(0.040)
Violent index	−0.101***	−0.097***	−0.098***
	(0.029)	(0.029)	(0.029)
Property index	−0.095**	−0.093**	−0.093**
	(0.044)	(0.044)	(0.044)

Notes: Each estimate comes from a separate regression of the logged adult arrest rate on an indicator for BBCE being in force, with $\text{stack} \times \text{agency}$ and $\text{stack} \times \text{year}$ fixed effects and controls for state economic conditions (unemployment and poverty rates, log minimum wage, log maximum TANF benefit for a family of three, state EITC rate, an indicator for a Democratic governor, log state population) and six contemporaneous SNAP administrative policies. Observations are weighted by agency covered population and standard errors in parentheses cluster on state, 44 clusters. Panel A replaces the arrest rate with police staffing from the FBI's Law Enforcement Officers Killed and Assaulted collection, so column (1) is the estimate, column (2) its standard error and column (3) the joint pre-trend p -value. Panel B adds staffing to the arrest regression: column (1) repeats the baseline on the agencies that report staffing, column (2) adds log sworn officers and column (3) adds sworn officers, total employees and the civilian share together. Staffing is an outcome of state fiscal conditions and moves with adoption, so conditioning on it risks conditioning on a post-treatment variable; Panel B bounds how much of the arrest effect could run through enforcement capacity rather than preferring that specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A2 Alternative estimators

The stacked design is used throughout for the reasons set out in Appendix A4: the comparison rule it implements is specific to each treated cohort and cannot be imposed by an estimator that pools all cohorts in one regression, and the stack-by-year fixed effect lets the time path differ across cohorts. The design is also a solution to the problem the recent literature identifies rather than a casualty of it, since every comparison agency in a stack is either never treated or adopts more than six years after the focal state, so no already-treated unit is ever used as a control. This appendix substantiates both claims: first by reporting what a plain two-way fixed effects specification would estimate on these data, then by varying the estimator while holding the sample and the comparison structure fixed.

What two-way fixed effects would estimate

Goodman-Bacon (2021) shows that with staggered adoption the two-way fixed effects coefficient is a weighted average of every two-group two-period difference-in-differences comparison available in the panel. Those comparisons fall into three types. A treated state may be compared with a state that never adopts. It may be compared with a state that adopts later, using the later adopter's pre-treatment period as the comparison. Or it may be compared with a state that adopted earlier, using the earlier adopter's post-treatment period as the comparison. Only the third type is problematic, and it is problematic precisely when treatment effects grow with exposure, because the comparison group is itself still responding to its own treatment.

To compute the decomposition I build a balanced state-year panel of the 41 sample states over 1996 through 2016, aggregated from the 2,342 sample agencies reporting in all twenty-one years, and decompose the unweighted two-way fixed effects estimate on the logged adult arrest rate. Table A5 reports the weight and average estimate for each comparison type, and the decomposition sums to the two-way fixed effects coefficient exactly.

The results are as the theory anticipates. Comparisons against never-treated states carry 49 percent of the weight and average -0.053 on the total index and -0.128 log points on the violent index. Comparisons using an already-treated state as the control carry 29 percent of the weight and average $+0.067$ and $+0.066$; they are positive in 26 of 36 individual cases on the total index. The pooled two-way fixed effects coefficients are $+0.006$ and -0.052 , so on the total index the contaminated comparisons erase the estimate entirely and flip its sign.

Figure A1 shows the pooled coefficient being assembled from those three contributions. On the total index the never-treated comparisons contribute -0.026 , the not-yet-treated comparisons add $+0.013$, and the already-treated comparisons add a further $+0.019$, landing on $+0.006$ against a stacked estimate of -0.087 . The property index is the starkest case: two-way fixed effects returns $+0.028$ where the stacked design returns -0.092 , so the comparisons the stacked design does not make overturn the entire effect. On the violent index, where the not-yet-treated comparisons happen to be negative, the contamination is milder and the two estimators are closer, -0.052 against -0.079 . Removing these comparisons is what the stacked design does by construction, and the distance between the grey bar and the dotted line in each panel is what that buys.

The imputation estimator

For a check that varies the estimator without varying anything else, I report the imputation estimator of Borusyak et al. (2024) on the same stacks the baseline uses. Stack-by-agency and stack-by-year effects and the state-year controls are fitted on untreated cells only, meaning comparison agencies at every event time and treated agencies before adoption. The counterfactual is then imputed for each of the 13,559 treated post-adoption cells and the estimate is the population-weighted mean of the residuals. Because the fitted effects are themselves estimated, inference must account for the first step; standard errors and percentile intervals come from 200 state block bootstrap draws in which states are resampled with replacement and the imputation is refitted inside each draw. Stack-by-year effects are held common across the resampled blocks, since a block-specific year cell would contain a single state and the state-year controls would be collinear with it.

Table 9 reports the result. The imputation estimates are 11.7, 9.3, and 12.7 log points against baseline estimates of 8.7, 7.9, and 9.2, and all three remain significant at the 10 percent level. The gap is a weighting difference rather than a disagreement about sign or timing: the imputation weights every treated cell by population, so it loads more heavily on the middle post-adoption years, where the estimated effect is deepest, whereas the stacked estimate averages across the whole post-adoption window. The baseline is therefore the more conservative of the two, and nothing in the paper's conclusions depends on which is preferred.

Table A5: Goodman–Bacon decomposition of the two-way fixed effects estimate

Outcome	TWFE	Average estimate [weight] by comparison type		
		Treated vs never	Earlier vs later	Later vs earlier
Total index	0.006	−0.053 [0.49]	0.059 [0.22]	0.067 [0.29]
Violent index	−0.052	−0.128 [0.49]	−0.037 [0.22]	0.066 [0.29]
Property index	0.028	−0.018 [0.49]	0.103 [0.22]	0.049 [0.29]

Notes: Goodman-Bacon (2021) shows that a two-way fixed effects coefficient is a weighted average of the two-by-two comparisons the panel makes available, so the reported TWFE estimate is exactly the sum of each type’s weight times its average estimate. The decomposition is computed on a balanced agency-year panel over 1996 to 2016. The last two columns use another adopting state as the comparison group; only the last of these uses a state that is already treated, and the stacked design excludes both by construction. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

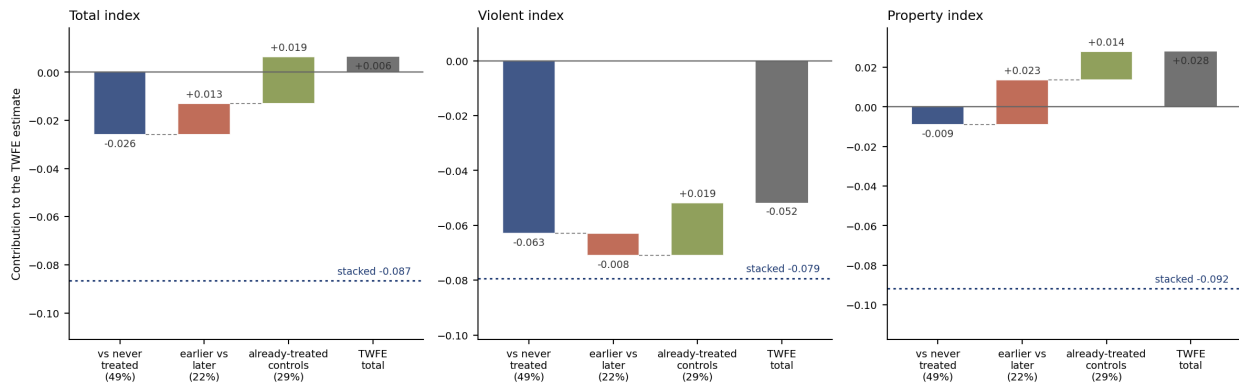


Figure A1: What the two-way fixed effects estimate is made of

Notes: The two-way fixed effects coefficient is a weighted average of the two-by-two comparisons the panel makes available, so it is exactly the sum of one contribution per comparison type, each equal to that type's weight in the estimator times its average two-by-two estimate. Bars show that sum being built and the grey bar is the total; percentages under each label are the weights. The dotted line is the stacked estimate, so the vertical distance between it and the grey bar is what the contaminated comparisons cost. Table A5 reports the number of comparisons and the average estimate within each type.

A3 The counterfactual in calendar time

Section 6.6 reports the counterfactual in event time, which is the scale the estimates are defined on. This appendix reports the same exercise summed by calendar year instead, which is the scale a policy audience works in and which shows how the totals accumulate.

Figure A2 plots it. Nothing is re-estimated: the event-time coefficient is removed from each treated agency-year exactly as before, and the resulting quantities are aggregated by year rather than by years since adoption. The totals at the end of the series are therefore the same 18.3 million recipient-years of participation and 211,931, 76,852 and 153,259 averted arrests reported in the body.

Two features of the figure need stating. The confidence bands widen sharply after 2009 because that is when most of the sample is inside its post-adoption window, so the cumulative total depends on more coefficients and inherits more of their covariance. And the decline in every series after 2015 is mechanical: states leave the calculation as they pass beyond their own six-year windows, not because the effect fades. Panel (c) reports point estimates only, since three overlapping delta-method bands are unreadable at that scale; panel (d) carries the uncertainty.

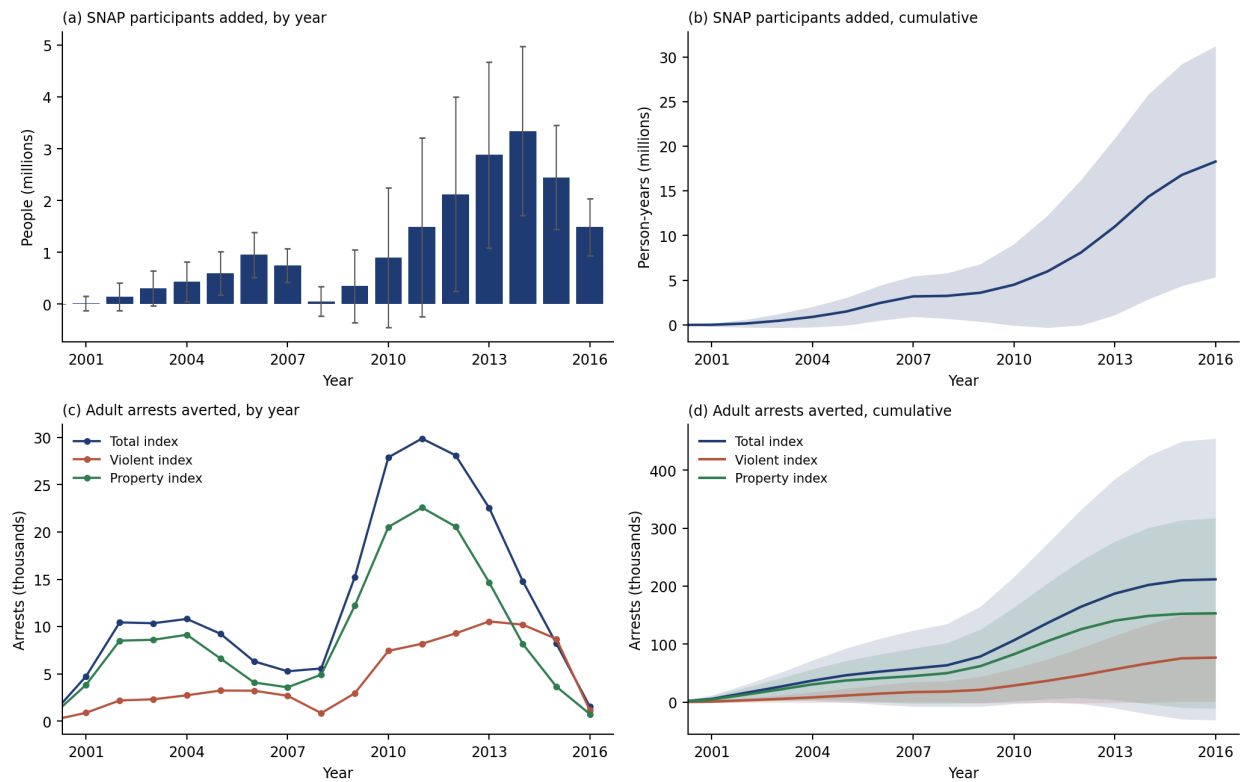


Figure A2: SNAP participation added and adult arrests averted, by calendar year

Notes: Annual and cumulative totals from the same counterfactual, summed by calendar year. Error bars in panel (a) and shaded bands in panels (b) and (d) are 95 percent intervals from the delta method using the full covariance matrix of the seven post-adoption coefficients. Panel (c) reports point estimates only, since three overlapping bands are unreadable; panel (d) carries the uncertainty. The decline after 2015 reflects states passing beyond their windows rather than a fading effect.

A4 Constructing the stacked data set

Notation

Let s index treated states and g_s the adoption year of state s , defined as the calendar year containing the first month in which BBCE is in force. Let i index police agencies and t calendar years. Write $k = t - g_s$ for event time.

Procedure

For each treated state the following steps are executed in order.

1. **Window.** Retain calendar years $t \in [g_s - 4, g_s + 6]$, eleven years in total. A state enters only if the entire window falls inside 1996 through 2016. This excludes the four states adopting in 2011 or later.
2. **Comparison pool.** Retain agencies in states that either never adopt BBCE within the sample period, or adopt more than six years after g_s . The six-year gap equals the post-adoption window, so no comparison agency is treated at any point during the focal state's window. Han and Wu (2026) impose the same rule with a three-year gap matching their three-year post window.
3. **Roster.** Within the pooled set of treated and comparison agencies, retain those reporting in all twelve months of every one of the eleven years and whose mean covered population over the window is at least 10,000. The resulting roster is fixed within the stack.
4. **Event time.** Define $D_{ist} = \mathbf{1}\{\text{agency } i \text{ in } s\} \times \mathbf{1}\{k \geq 0\}$. Lead and lag indicators $\mathbf{1}\{k = j\}$ are interacted with treated status for $j \in \{-4, \dots, -2\} \cup \{0, \dots, 6\}$, with $k = -1$ omitted.
5. **Stack identifier.** Label every observation with s and append it to the pooled data set.

A state is dropped if no agency survives step 3. Because a comparison agency may serve several stacks, the 2,784 distinct agencies generate 255,640 stack-agency-year observations.

Table A6: Stack composition

State	Cohort	Adoption month	Lever	Agencies	Treated	Comparison states	Reversal
DE	2000	February	Income and asset	1,221	4	29	
ME	2000	September	Income and asset	1,237	20	29	
MI	2000	October	Income and asset	1,328	111	29	
OR	2000	December	Income and asset	1,247	30	29	
MA	2001	November	Income and asset	1,294	51	28	
MD	2001	March	Income and asset	1,274	31	28	
SC	2001	April	Income and asset	1,284	41	28	
TX	2001	September	Income and asset	1,451	208	28	
WA	2004	May	Asset only	516	64	12	
MN	2006	December	Asset only	505	65	10	
AZ	2007	June	Income and asset	490	28	10	
GA	2008	March	Asset only	487	15	10	
NY	2008	January	Asset only	571	99	10	
OH	2008	October	Asset only	548	76	10	
PA	2008	October	Asset only	691	219	10	
WV	2008	October	Asset only	485	13	10	
CA	2009	July	Asset only	864	387	10	
CT	2009	July	Income and asset	539	62	10	
ID	2009	June	Asset only	507	30	10	
NH	2009	May	Income and asset	501	24	10	
NV	2009	April	Income and asset	487	10	10	
OK	2009	June	Asset only	507	30	10	
RI	2009	April	Income and asset	505	28	10	
AL	2010	February	Asset only	491	1	10	
HI	2010	October	Income and asset	492	2	10	
IL	2010	March	Asset only	491	1	10	
KY	2010	June	Asset only	496	6	10	
LA	2010	June	Asset only	500	10	10	2014
MS	2010	June	Asset only	498	8	10	2019
NC	2010	July	Income and asset	530	40	10	
NJ	2010	April	Income and asset	710	220	10	
NM	2010	April	Income and asset	493	3	10	

Notes: One row per stack. The cohort is the calendar year containing the adoption month, so the adoption year is only partly treated; Table A1 reports the estimate under fractional coding of that year. Comparison states are those never adopting or adopting more than six years after the focal state. Louisiana's 2014 reversal falls inside its window and Mississippi's 2019 reversal falls outside the sample period. The sample is 2,784 agencies in 44 states, 32 stacks, 255,640 stack-agency-year observations.

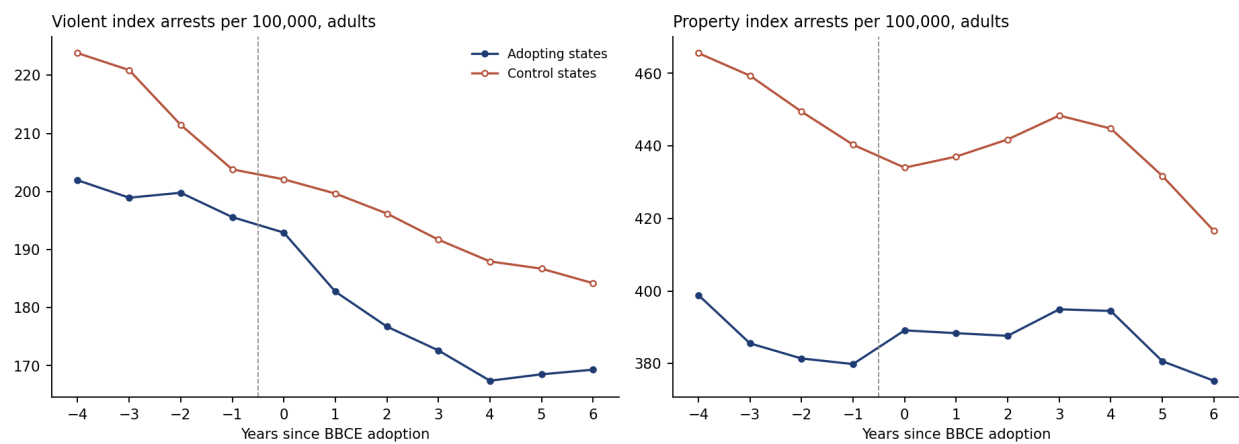


Figure A3: Raw arrest rates by event time

Notes: Population-weighted mean arrest rates by years since adoption, computed separately for treated and comparison agencies within the stacked sample, without controls or fixed effects.

A5 Data and measurement

Table A7: Variables, definitions, and sources

Variable	Definition	Source
<i>Panel A. Outcomes</i>		
Adult arrests, by offense	Arrests of persons 18 and over, summed over the twelve adult age-by-sex cells, per 100,000 residents covered by the agency, in logs	FBI, Arrests by Age, Sex, and Race, annual ICPSR releases 1996–2019 (Federal Bureau of Investigation, 1996–2019)
Violent index	Murder and nonnegligent manslaughter, forcible rape, robbery, aggravated assault	ASR
Property index	Burglary, larceny, motor vehicle theft; arson excluded	ASR
Total index	Union of the violent and property indices	ASR
Arrestee age, sex, race	Age groups 18–24 and 25–54; sex; White and Black adult arrestees	ASR
Agency covered population	Largest monthly covered population reported in the year; the denominator of every rate and the regression weight	ASR
<i>Panel B. Treatment</i>		
BBCE in force	Indicator equal to one from the calendar year containing the first month in which BBCE is in force, and one thereafter	USDA Economic Research Service, SNAP Policy Database (state-month)
Gross income limit	Income eligibility threshold as a percent of the federal poverty guideline, at adoption	SNAP Policy Database
Asset-test mechanism	Whether the state eliminated or relaxed the asset test, and the lever classification used in Section 6.4	SNAP Policy Database; classification from Han and Wu (2026)
<i>Panel C. First stage</i>		
SNAP recipients, caseloads	State-year individual recipients and household caseloads, in logs and per capita	UKCPR National Welfare Data (University of Kentucky Center for Poverty Research, 2023), compiled from USDA Food and Nutrition Service administrative counts
<i>Panel D. State controls</i>		
Unemployment rate, poverty rate	State annual rates	UKCPR National Welfare Data (University of Kentucky Center for Poverty Research, 2023)
Minimum wage	Higher of the state and federal minimum wage, in logs	UKCPR
Maximum TANF benefit	Maximum monthly benefit for a family of three, in logs	UKCPR
State EITC rate	State credit as a percent of the federal credit	UKCPR
Democratic governor	Indicator for the governor's party	UKCPR
State population	In logs	UKCPR
Six SNAP administrative policies	Simplified reporting, combined application project, vehicle exclusions, EBT issuance, online application, short recertification periods	USDA Economic Research Service, SNAP Policy Database
<i>Panel E. Used only in Appendix A1</i>		
Sworn officers, civilians, total employees	Full-time sworn officers, civilians, and total employees per 100,000 covered residents	FBI, Law Enforcement Officers Killed and Assaulted, concatenated by Kaplan (2021)
Offenses known to police	Part I offenses recorded by the agency before any arrest decision; aggravated assault is net of simple assault	FBI, Offenses Known and Clearances by Arrest, ICPSR 100707

Notes: The arrest panel is built from the twenty-four annual ICPSR releases of the Arrests by Age, Sex, and Race collection, whose records are at the agency-month-offense level; the individual study numbers appear in the bibliography entry for Federal Bureau of Investigation (1996–2019). Panel E lists series that enter no specification reported in the body.

Table A8: Crime outcomes, definitions, and FBI offense codes

Outcome	FBI code	Definition
<i>Panel A. Composite indices</i>		
Total index	01A-07	Sum of the violent and property indices
Violent index	01A-04	Murder and nonnegligent manslaughter, forcible rape, robbery, aggravated assault
Property index	05-07	Burglary, larceny, motor vehicle theft
<i>Panel B. Violent component offenses</i>		
Murder	01A	Willful killing of one human being by another; excludes negligent manslaughter, traffic fatalities, attempts, and justifiable homicide
Rape	02	Carnal knowledge of a female forcibly and against her will, and attempts, under the legacy definition in force through 2012; the revised definition applies from 2013
Robbery	03	Taking or attempting to take anything of value from a person by force, threat of force, or by putting the victim in fear
Aggravated assault	04	Unlawful attack for the purpose of inflicting severe bodily injury, usually involving a weapon; simple assault is a separate code
<i>Panel C. Property component offenses</i>		
Burglary	05	Unlawful entry of a structure to commit a felony or theft, including attempts
Larceny	06	Unlawful taking of property from the possession of another without force, violence, or fraud; includes shoplifting and theft from vehicles
Motor vehicle theft	07	Theft or attempted theft of a self-propelled vehicle that runs on the surface and not on rails
<i>Panel D. Reported but outside the indices</i>		
Simple assault	08	Assault without a weapon and not resulting in serious injury; a Part II offense, reported for reference only
Arson	09	Willful burning of property; excluded from the property index because reporting coverage is incomplete

Notes: Codes are those of the Arrests by Age, Sex, and Race collection (Federal Bureau of Investigation, 1996–2019). Each outcome is the sum of adult arrests across its codes at the agency-year level, expressed per 100,000 residents covered by the agency and entered in logs. The FBI revised the rape definition in 2013 to remove the force and female-victim requirements, so the rape series is not fully comparable across that break; the change is national and is absorbed by the stack-by-year effects.